

GENERAL THETA FOUNDATIONS I: INTRINSIC CONTINUATION GEOMETRY AND CAUSAL MEMORY

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ABSTRACT. We study the number of persistent states needed to realize a causal statistical response. For a complete finite response array we define an intrinsic support-component rank from all normalized continuations. Every stochastic realization obeys this cutwise lower bound. If the convex hull in each component is a simplex, residual vertices attain all these bounds simultaneously through normalized causal updates. The criterion is finite and effective on rational data, and the complete class is closed under independent products of terminal channels, with multiplicative state counts. A growing binary random-access experiment then separates predictive dimension from positive memory: its exact minimum whole-schedule peak is $n + 1$ for $0 < \eta \leq n^{-2}$ and 2^n for $1 - n^{-1} < \eta \leq 1$, even when acquisition order is chosen adaptively. The weak-signal bound is attained by an explicit sequence of compatible enclosing simplices, not merely by a final encoder. Minimax contact connects the classification to the marked physical audit and its exact five-state decision law. The previously established positive-noise saddle, fixed-clock validation-order classification, support-boundary jump, causal foundations and transport proofs are retained as supporting mathematics. Exact physical counts assume a specified target and atomic stochastic rows; acquired calibration has separately priced approximation guarantees.

CONTENTS

1. Introduction	3
1.1. Resource and parameter conventions	4
1.2. Organization and the retained development	4
2. An intrinsic continuation invariant	4
2.1. Normalized causal residuals	4
2.2. Exact schedule fields without selected certificates	6
2.3. Why the completeness hypothesis has content	7
3. Products and intrinsic memory growth	7
4. A scalable full-support experiment with two exact memory laws	8
4.1. The intrinsic checkpoint geometry	9
4.2. A compatible sequence of enclosing simplices	9
4.3. A geometry stronger than rank	10
5. From saddle contact to intrinsic memory	11
5.1. The marked decision geometry as a complete instance	12
5.2. Known-target physical exactness and acquisition	12
6. Theorem-level relation to realization and coding	13
Supporting foundations and physical proofs	15
Appendix A.1. Causal experiments and their predictive quotients	15
A.1.1. The experimental object	15

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A.1.2. Prepared predictive equivalence	16
Appendix A.2. Attainable resolution and resource-aware simulation	17
A.2.1. Checkpoint and online prediction	17
A.2.2. Causal morphisms and simultaneous error budgets	18
Appendix A.3. Experiments, architectures and attainable resources	20
Appendix A.4. A statistical dual that is uniform over architectures	21
Appendix A.5. Controlled experiments without a dominating action	22
A.5.1. An informative action choice	24
Appendix A.6. Controlled experiments at a prescribed memory profile	25
A.6.1. A parameterwise occupation state	25
A.6.2. An exact informative-control bottleneck	27
Appendix A.7. Saddle faces and causal realization	28
A.7.1. A normal form over a finite causal interface	28
A.7.2. The realization theorem	30
A.7.3. Faces as memory witnesses	31
Appendix A.8. Facial certificates for schedule-optimized memory	32
Appendix A.9. Exact minimax laws for a marked product family	34
A.9.1. Complementary slackness and the event alphabet	35
Appendix A.10. A self-contained collision interface	36
A.10.1. Preparation and elastic scattering	36
A.10.2. Positive noise and the actual mark	37
Appendix A.11. The physical experiment and its sharp preparation boundary	38
A.11.1. An exact rational certificate against one preparation	39
A.11.2. A marked five-event selector	40
A.11.3. Causal compilation and the full register profile	40
Appendix A.12. The exact two-preparation marked saddle	42
Appendix A.13. A continuous two-preparation marked family	45
Appendix A.14. Exact memory from a marked saddle	46
A.14.1. The sharp serial-validation peak	48
Appendix A.15. A full-support marked saddle	50
Appendix A.16. Exact statistical form of the Brownian collision sensor	52
Appendix A.17. Optimal validation orders and a positive-noise memory jump	54
Appendix A.18. Preparation limits and localization of least-favorable priors	57
Appendix A.19. An exact preparation–decision-memory law	59
Appendix A.20. Integer memory and the failure of a naive partition formula	61
Appendix A.21. Quantitative certificates for a finite compatible architecture	63
Appendix A.22. Autonomous noisy testing with a precision and time ledger	64
Appendix A.23. Physical consumers and the first-principles connection	66
Appendix A.24. A sharpened physical consumer and its simulation budget	67
Appendix A.25. Dependencies, retained results and remaining distinctions	68
References	68

1. INTRODUCTION

A finite stochastic experiment can have a low-dimensional predictive image and nevertheless require many persistent states. The reason is positivity: an internal state describes a normalized future experiment, not an arbitrary vector in a linear span. Causality imposes a second constraint. Positive representations at successive cuts must be joined by common stochastic updates, and a schedule chosen from observations cannot reappear as an uncharged clock.

We seek conditions under which those constraints admit an intrinsic and exact state count. A lower bound obtained from selected faces and an independently supplied upper program can certify a particular example. It does not yet tell us when the continuation geometry itself determines a complete answer. Theorem 2.2 supplies such a condition for a finite class of causal experiments. Its data are all normalized continuation rows of the response array, their minimal faces in the causal polytope, and the connected components of the face-intersection graph. The resulting sum of affine ranks is a lower bound against every stochastic realization. If the residual convex hull in each component is a simplex, a canonical residual-vertex construction attains those ranks at every cut in one machine.

The condition does not assume a dimension-growth law, a metric entropy exponent, or a proposed controller. It can be checked by exact linear feasibility and rank. It is also not universal: an explicitly given full-support square has rank three and needs four states. The failure example makes clear what the completeness hypothesis excludes. For independent terminal channels the complete class is closed under products, and its intrinsic capacities multiply (Theorem 3.1). At every declared schedule vertex this gives an exact field rather than a field of analyst-selected certificates. The schedule optimum follows by a bottleneck recurrence.

A second theorem gives a more substantial separation in a growing experiment. Read n bits, then answer a later coordinate query with specified bias η . Every input and every query must have exactly the required conditional output law. The checkpoint geometry is the containment of $[-\eta, \eta]^n$ in a convex hull inside $[-1, 1]^n$. For $0 < \eta \leq n^{-2}$, a sequence of enclosing simplices has $(t+1)$ vertices after t reports and is closed under appending the next bit. Its barycentric updates implement the channel with peak $n+1$, which the affine dimension forces. For $\eta > 1 - n^{-1}$, disjoint corner caps force 2^n states, attained by full retention. Both lower bounds hold after any adaptive acquisition schedule, because the whole acquired transcript must pass through the state before the query. Both regimes include full-support laws. Thus the same predictive rank $n+1$ can be attained by causal memory or be exponentially smaller than it.

The statistical application enters in the opposite direction: a minimax saddle first forces continuations, and the intrinsic geometry then prices them. Bayes indifference alone does not specify an optimal continuation. For the marked two-preparation family, contact and stationarity fix two coordinates left free by a least-favorable prior. The five forced rows have canonical support components of dimensions 1, 0, 1, recovering the exact count $2 + 1 + 2 = 5$. These rows remain forced in the full-support physical rectangle. Their independent query-channel products have capacity 5^m ; this is a statement about specified product continuations, not a new formula for the multi-block minimax value.

The proof structure is therefore

$$\begin{aligned}
 & \text{uniform saddle contact} \longrightarrow \text{forced normalized continuations} \\
 & \longrightarrow \text{intrinsic positive geometry} \longrightarrow \text{compatible simplex updates} \\
 & \longrightarrow \text{one causal realization with matching memory.}
 \end{aligned}$$

The general positive-realization and residual-automaton mechanisms have classical antecedents. Section 6 compares the precise claims with stochastic realization, residual automata, nested-polytope factorization, incomplete-machine minimization and random-access coding. We do not claim a new general minimax interchange or a solution of unrestricted positive-rank minimization.

1.1. Resource and parameter conventions. All persistent randomness belongs to the register. A fixed finite program and known row constants are read-only; an unknown target is not a free constant. The finite-state theorems use atomic stochastic rows, not free finite-length exact sampling of arbitrary real coins. The random-access query is one atomic input, explicitly distinct from a bit-serial query buffer. Input-dependent acquisition choices are allowed, but their retained protocol information is charged. The physical twelve-state optimum is over externally declared validation clocks; no adaptive collision optimum is inferred from it.

For a specified positive-noise Brownian target the exact physical saddle and five/twelve-state conclusions are retained. Acquired target calibration instead gives an approximate-score controller with finite counter and precision costs. The main theorems keep these quantifiers separate.

1.2. Organization and the retained development. Sections 2–6 contain the current theorem spine and its proofs. The supporting sections then retain the previous causal foundations, exact saddles, physical identification and validation-order proofs, revelation laws, finite-architecture certification, all-preparation localization, and transport consumers. Their stable theorem labels remain resolvable in this single manuscript. No predecessor proof is discarded to shorten the main narrative. The historical A2 geometric root and B4/C2 operator gates retain their own hypotheses; neither the present class theorem nor the growing query example replaces those independent derivations.

2. AN INTRINSIC CONTINUATION INVARIANT

We first give a finite realization theorem whose input is the complete response law, rather than a selected collection of lower certificates. The distinction between a law and an implementation is essential: states in a competing implementation need not themselves be residuals of the law. The lower bound below allows all such states.

2.1. Normalized causal residuals. Fix a finite horizon, finite alphabets, and a list of charged cuts. At an elementary stage the machine chooses an action a , then receives a report y . A terminal output is an action followed by a dummy report. Let $C(\mathbf{a} \mid \mathbf{y})$ be a nonnegative normalized array whose action-prefix marginal depends only on the reports preceding that prefix. Thus an action cannot read its own future report. Every necessary buffer cut is an elementary cut. Time-dependent rows are allowed; a persistent seed, data-dependent clock, or protocol register is not a free input.

These conventions describe exact realization of a *specified* causal array on the full declared alphabet. For a model imposing only almost-sure requirements, one must first specify a legal completion on unsupported words, or optimize over those completions. Changing the completion can change the answer. Additional autonomous row-identification constraints are not part of the clocked theorem below.

For a positive-mass controller prefix $h = (\mathbf{a}_{\leq t}, \mathbf{y}_{\leq t})$, let R_h be the normalized array of future actions conditional on that prefix and on externally supplied future reports. Positivity of the prefix refers to its marginal in C , with past reports fixed, and not to a prior on environmental parameters. Write $\mathcal{R}_t$ for the finite set of distinct such arrays. Duplicate arrays are removed. Let $\mathcal{P}_t$ be the polytope of all normalized causal arrays on the remaining interface. Its defining equations express normalization and nonanticipation; its inequalities express nonnegativity.

Legality restrictions are included in the response array's zero pattern. A mandatory retained output or buffer is included through its prescribed later readout, so that deleting it changes the continuation task.

If $R \in \mathcal{R}_t$ and its next action has probability $p_R(a) > 0$, then, for every next report y ,

$$(1) \quad R(a\mathbf{a} \mid y\mathbf{y}) = p_R(a)R^{a,y}(\mathbf{a} \mid \mathbf{y}), \quad R^{a,y} \in \mathcal{R}_{t+1}.$$

The scalar $p_R(a)$ is independent of y . If $p_R(a) = 0$, the entire left side is zero. These are normalized residuals of a controller, not posteriors in its runtime register.

Definition 2.1 (Support components and their rank). For $R \in \mathcal{R}_t$, denote its minimal face in $\mathcal{P}_t$ by $F_t(R)$. Join two residuals when their minimal faces intersect. Let $\mathcal{R}_{t,1}, \dots, \mathcal{R}_{t,c_t}$ be the connected components of this graph, and put

$$(2) \quad d_{t,j} = \dim \text{aff } \mathcal{R}_{t,j}, \quad s_t(C) = \sum_{j=1}^{c_t} (d_{t,j} + 1), \quad e_t(C) = \sum_{j=1}^{c_t} |\text{ext conv } \mathcal{R}_{t,j}|.$$

Call the residual geometry *componentwise simplicial* at t if each $\text{conv } \mathcal{R}_{t,j}$ is a simplex, including a point as a zero-dimensional simplex.

Everything in this definition is determined by C and the charged interface. For finite rational data, face intersection is a linear feasibility problem with the two zero patterns imposed, affine dimension is a matrix rank, and extreme-point membership is another linear feasibility problem. No witness subfamily or trial controller enters the definition. Since normalized residuals lie in an affine hyperplane not containing zero, $d_{t,j} + 1$ equals their ordinary row rank. For terminal-output channels the ambient polytope is a product of probability simplices; two minimal faces intersect precisely when their allowed-output sets intersect at every future input.

Theorem 2.2 (Intrinsic simultaneous realization). *Every stochastic clocked realization of C , with all persistent variables included in its register, has*

$$(3) \quad K_t \geq s_t(C)$$

at every charged cut. A single stochastic realization has profile $K_t = e_t(C)$ simultaneously at all cuts. If the residual geometry is componentwise simplicial at every cut, then

$$(4) \quad K_t^{\min} = s_t(C) = e_t(C) \quad \text{simultaneously,} \quad W^{\min}(C) = \max_t s_t(C).$$

A realizing machine is obtained directly from residual vertices and barycentric one-step updates. Thus the statement determines an exact profile, not just unrelated lower bounds at individual cuts.

Proof. Conditioning a realizing machine on a controller prefix gives

$$(5) \quad R_h = \sum_s E_h(s)B_{t,s}, \quad E_h(s) \geq 0, \quad \sum_s E_h(s) = 1, \quad B_{t,s} \in \mathcal{P}_t.$$

Here $B_{t,s}$ is the future program starting in the actual state s . It may lie outside the convex hull of the residuals. If $E_h(s) > 0$, the defining property of a face implies $B_{t,s} \in F_t(R_h)$. A state used for residuals in two different components would therefore belong to two intersecting minimal faces, which would join those components. This is impossible. The states used by distinct components are disjoint. Within component j , the affine span of its residuals is contained in the affine span of its used state continuations, and hence at least $d_{t,j} + 1$ states are required. Summing proves (3). This argument does not replace stochastic encoders by deterministic ones.

For the upper construction, at cut t choose as states all extreme points of the component convex hulls. Every such point is an actual residual, since the residual set is finite. Each residual is a convex combination of these chosen states. Given a state R , choose its next action with

probability $p_R(a)$. For positive $p_R(a)$, represent $R^{a,y}$ from (1) as a convex combination of the chosen states at cut $t + 1$, and use those coefficients as the update row after report y . At zero $p_R(a)$ choose an arbitrary update on the unused row. The action row remains independent of the unread report. Initialization is the decomposition of C at the first cut; the terminal continuation is realized by its prescribed output row.

Backward induction proves the entire future array from every chosen state. In particular, the updates at different cuts implement the same law; no separate factorization is being spliced without checking its shifts. This gives the profile $e_t(C)$. In the simplicial case each component has exactly $d_{t,j} + 1$ vertices, so the upper and lower profiles coincide. Within each component the barycentric coefficients are unique, although arbitrary minimum-size realizations need not be unique or have states equal to residuals. $\square$

Corollary 2.3 (A complete, finite test for the stated class). *For a complete finite rational response array, componentwise simpliciality and the numbers in (4) can be decided from linear systems, rational linear feasibility, and a finite graph. When the test succeeds, rational transition tables realizing the exact profile can be constructed. The same holds for real algebraic input using exact sign and linear-feasibility decisions over the specified ordered field.*

Proof. Enumerate the positive-prefix normalized residuals and their zero patterns. Solve the pairwise face-intersection problems in the stated causal polytope, find the connected components, and find the extreme residuals of each component hull. A component hull is a simplex exactly when its number of vertices equals its affine dimension plus one. Solve the barycentric systems in the proof of Theorem 2.2. All operations are finite and preserve the indicated field. $\square$

The input table can be exponentially large in the horizon. The corollary is a completeness and synthesis result for an explicit class, not a polynomial algorithm in a succinct program length. It does not say that ordinary rank solves unrestricted positive realization. Nor does it claim that residual-state minimization, by itself, minimizes over all probabilistic machines; that last conclusion uses the separated-component lower bound. The distinction is also present in residual automata [21] and nested-polytope formulations [19].

2.2. Exact schedule fields without selected certificates. Consider a terminal-output channel on finitely many tapes, internally ordered on each tape, with a full product input alphabet. At a vector v of consumed-prefix lengths let $\mathcal{R}_v$ consist of the channel rows for *all* prefixes at v as functions of the remaining suffixes. The rows and their minimal faces are independent of the path used to reach v . Define $s(v), e(v)$ by (2).

Corollary 2.4 (Intrinsic declared-order minimization). *If every component hull at every schedule vertex is a simplex, the least width on each declared schedule π is $\max_{v \in \pi} s(v)$. In particular,*

$$(6) \quad W_{\text{clk}}^{\min} = \min_{\pi} \max_{v \in \pi} s(v).$$

The minimum is computed by the bottleneck recurrence on the finite schedule graph. The same residual-vertex construction supplies the updates for every allowed next-tape restriction.

Proof. Restriction of an actual prefix row by any next tape symbol is an actual row at the corresponding child vertex. Apply the proof of Theorem 2.2 on the entire schedule graph, then on each path. The last-edge decomposition of paths gives the bottleneck recurrence. No lower certificate or upper residual family is an extra input to this corollary. $\square$

An observation-dependent choice of tape has to be included as a controller action in C . Theorem 2.2 also classifies such a specified causal array when its normalized residual geometry passes the same test. Equation (6) alone optimizes declared paths, not all adaptive arrays. A

scalable experiment for which the peak optimum is unchanged even by adaptive acquisition is solved in Section 4.

2.3. Why the completeness hypothesis has content. The condition is neither universal nor a restatement of the answer. Consider four binary-query channels with response vectors $(\epsilon, \epsilon), (\epsilon, 1 - \epsilon), (1 - \epsilon, \epsilon), (1 - \epsilon, 1 - \epsilon)$, where $0 < \epsilon < 1/4$. All output probabilities are positive, so there is one support component. Its affine dimension is two and $s = 3$, but four states are necessary. Indeed, for each corner of the square a convex representation of its displayed row needs a generator with total coordinate error at most $2\epsilon < 1/2$. Two distinct corner-error sums add to at least one, so no generator can serve two corners. Four deterministic corner generators suffice. The nonsimplicial square therefore has state count four rather than three. This is a strict failure of the lower bound outside its completeness class, not an unsupported claim that the class contains every experiment.

3. PRODUCTS AND INTRINSIC MEMORY GROWTH

A classification becomes more useful when it survives independent composition. For the following statement a continuation is a normalized channel from a finite future-query alphabet Y to a finite output alphabet O . Multiple components are allowed. The joint experiment requires the whole product output law, not merely the correct marginal means.

Theorem 3.1 (Multiplicativity on the complete class). *Let $\mathcal{R} \subset \prod_{y \in Y} \Delta(O)$ and $\mathcal{S} \subset \prod_{z \in Z} \Delta(A)$ be finite sets whose support-component hulls are simplices. Form the independent product rows*

$$(R \otimes S)(o, a \mid y, z) = R(o \mid y)S(a \mid z).$$

Their component hulls are again simplices, and their intrinsic capacities satisfy

$$(7) \quad s(\mathcal{R} \otimes \mathcal{S}) = s(\mathcal{R})s(\mathcal{S}).$$

Consequently every joint stochastic encoder and decoder for the product requires this many states; correlated internal implementations cannot reduce the number. For independent terminal channels whose residual sets pass the test at every individual cut, the product profile on any interleaving is exactly the product of the individual capacities at the corresponding vector of cuts.

Proof. Minimal faces of a query channel are given by the allowed outcomes at each query. Two product faces intersect exactly when the faces in each factor intersect: supports multiply, and a nonempty intersection must have nonempty projected supports in both factors for every query. The component graph of the product is the graph product with a loop at each vertex. Its connected components are precisely products of components, because one can move along a path in one factor while holding the other fixed.

In a component with k simplex vertices, the flattened normalized channel vectors of those vertices are linearly independent. Indeed, normalization turns any linear dependence into an affine dependence. For components with k and l vertices, their kl product vectors are linearly independent by the tensor-product rank identity. Every product of rows is a convex combination of these vectors, and every one of these product vertices is itself a product row. Hence the component hull is their $(kl - 1)$ -simplex. Summing kl over pairs of components gives (7). The cut lower bound allows arbitrary normalized joint generators, not just product generators. The upper construction uses products of the individual generators. Restriction of a product residual changes only the factor whose input is read, so compatibility holds simultaneously throughout the interleaving graph. $\square$

The independence in the specification of the output channel is indispensable. Independent inputs or additive payoffs alone do not force a product of optimal response channels. In particular, (7) is not a multiplicativity claim for arbitrary nonnegative rank or for the marked minimax values U_N .

Corollary 3.2 (A full-support family at a preparation barrier). *For $q \geq 2$ and $0 < \lambda < 1$, let*

$$T_\lambda(b | a) = \lambda \mathbf{1}_{a=b} + (1 - \lambda)/q, \quad a, b \in [q].$$

Acquire n input symbols before releasing any output; subsequently produce the specified independent product channel $T_\lambda^{\otimes n}$. The exact minimum peak is q^n , even if acquisition and output schedules depend on retained observations and internal randomization, provided all schedule memory is charged and no past transcript can be read back. For a declared schedule, at quiescent cuts between completed release operations, the exact width with m acquired but unreleased symbols is q^m . An intermediate held-output cut is charged separately using the active count immediately before that release. At $\lambda = 0$, one state suffices until an output-buffer cut.

Proof. T_λ has eigenvalues 1 and λ with multiplicity $q - 1$, so its q rows are independent and form a simplex. All entries are positive. Theorem 3.1 gives q^m at any fixed active cut. For an adaptive schedule at the barrier, condition on the complete input word x . Its state law is an encoder $E(s | x)$; the law of the entire subsequently produced labelled output vector is a decoder $D(\mathbf{b} | s)$. Any output-order choices are included in this law before taking the labelled-output projection. Thus $T_\lambda^{\otimes n} = ED$, whose rank q^n bounds the state count. This argument charges, rather than reveals, the acquisition transcript. Retaining the full input word gives the matching fixed-order upper bound. Releasing a symbol samples its output and discards that input; even if the output itself is held for one cut, there are at most $q q^{m-1} = q^m$ states. At $\lambda = 0$ the outputs are independent of all inputs, and can be sampled without retaining them. $\square$

This is a rank-boundary phenomenon entirely inside full support, unlike the physical support boundary in Theorem A.17.2. For every fixed n , the product law tends uniformly in total variation to the input-independent output law, with defect at most $n\lambda(1 - 1/q)$. Thus a one-state generator approximates it with that defect, while every nonzero λ has the stated exact memory. The estimate follows by coupling each output with an independent uniform symbol and summing the per-coordinate disagreements. Exact memory and approximate statistical performance are different quantities.

4. A SCALABLE FULL-SUPPORT EXPERIMENT WITH TWO EXACT MEMORY LAWS

We next keep the final answer alphabet binary while the preparation length grows. This distinguishes intrinsic positive memory from ordinary predictive dimension, and gives an exact causal construction rather than a checkpoint factorization alone.

An input $x = (x_1, \dots, x_n) \in \{-1, 1\}^n$ is fixed. Each coordinate can be read once during acquisition. After acquisition an external query $j \in [n]$ arrives as one atomic symbol, and a bit $B \in \{-1, 1\}$ is emitted. For a specified $0 \leq \eta \leq 1$, the required experiment is

$$(8) \quad \Pr(B = b | x, j) = \frac{1 + \eta x_j}{2}.$$

Acquisition may choose the next unread coordinate using the retained state. At the query cut the entire acquisition transcript has been discarded except for that state. Neither a public persistent seed nor a transcript of the chosen order is given to the decoder. The external round number is a declared clock; any data-dependent schedule variables must be stored. The output is sampled in the update receiving j , so j need not be buffered after that update. Exact arbitrary stochastic rows are atomic, as in the preceding realization theorems.

Let $W_n(\eta)$ be the least whole-schedule peak over these controllers. Let $K_n(\eta)$ be the least number of states just before the query, without an upstream cap. For $0 < \eta < 1$ every required answer probability has full support. Equality with (8) is required for every input and every query; an average success criterion is different.

Theorem 4.1 (Linear and exponential exact memory regimes). *For every integer $n \geq 1$,*

$$(9) \quad K_n(\eta) = W_n(\eta) = n + 1, \quad 0 < \eta \leq n^{-2},$$

$$(10) \quad K_n(\eta) = W_n(\eta) = 2^n, \quad 1 - n^{-1} < \eta \leq 1.$$

In the weak-signal regime a fixed-order streaming controller has exactly $t + 1$ states after t input bits, simultaneously for $0 \leq t \leq n$. For that fixed order these counts are minimal. Both peak lower bounds hold for arbitrary adaptive acquisition schedules. The two displayed regimes are not asserted to exhaust the intermediate parameter range.

4.1. The intrinsic checkpoint geometry. For a query-state s , write $z_s(j) = \mathbb{E}[B \mid s, j] \in [-1, 1]$. An encoder gives (8) exactly if and only if

$$(11) \quad \eta x = \sum_s E(s \mid x) z_s \quad \text{for every } x \in \{-1, 1\}^n.$$

Thus $K_n(\eta)$ equals the smallest number of points in $[-1, 1]^n$ whose convex hull contains $[-\eta, \eta]^n$. Necessity is conditioning on the state. Conversely any such containment supplies a stochastic encoder and binary decoder. This is a normalized nested-polytope formulation, not an identification with unrestricted matrix rank [19]. It says nothing by itself about the memory needed to compute the encoder online.

For $\eta > 0$, the inner cube has affine dimension n , whence

$$(12) \quad K_n(\eta) \geq n + 1.$$

Equivalently, the matrix in (8), with input words as rows and pairs (j, b) as columns, has ordinary rank $n + 1$. Adaptive acquisition cannot evade (11): after the barrier there is still only one retained state law for each complete input, and the same decoder on every input.

4.2. A compatible sequence of enclosing simplices. We construct the encoder using only a label, never a stored real vector. For $t \geq 1$ put $b_t = t\eta$ and define

$$(13) \quad P_t = \{z \in \mathbb{R}^t : z_i \geq -b_t, \sum_{i=1}^t z_i \leq b_t\}.$$

Its vertices are $v_{t,0} = (-b_t, \dots, -b_t)$ and $v_{t,i} = v_{t,0} + (t+1)b_t e_i$, $1 \leq i \leq t$. The distinguished coordinate of $v_{t,i}$ is $t^2\eta$ and its other coordinates are $-t\eta$. Therefore $P_t \subset [-1, 1]^t$ when $\eta \leq n^{-2}$ and $t \leq n$. Set $P_0 = \{()\}$. The key causal inclusion is

$$(14) \quad P_t \times \{-\eta, \eta\} \subset P_{t+1}.$$

Indeed old coordinates are at least $-t\eta$, the new coordinate is at least $-\eta$, and the new sum is at most $t\eta + \eta = b_{t+1}$. For $w \in P_t$ its barycentric coordinates are

$$(15) \quad \lambda_0(w) = \frac{b_t - \sum_i w_i}{(t+1)b_t}, \quad \lambda_i(w) = \frac{w_i + b_t}{(t+1)b_t} \quad (1 \leq i \leq t).$$

They are nonnegative, sum to one, and reconstruct w .

Proof of (9). At time t the register is an index $s \in \{0, \dots, t\}$. After bit x_{t+1} , replace it by index s' sampled from the barycentric coordinates of $(v_{t,s}, \eta x_{t+1})$ in P_{t+1} . These are legal rows by (14). Induction gives

$$\sum_s \Pr(s \mid x_1, \dots, x_t) v_{t,s} = \eta(x_1, \dots, x_t).$$

Given the final state s and query j , draw the answer with mean $v_{n,s}(j)$. This is a valid binary row because $P_n \subset [-1, 1]^n$. The preceding identity proves (8) exactly. It also proves the entire online implementation, not just existence of a final stochastic matrix. The real vectors in the proof are read-only row descriptions; the machine stores only their indices. No past coin, real accumulator, or chosen-input transcript survives outside the register. Formula (15) shows that all acquisition rows are rational and independent of $\eta > 0$, because that common factor cancels. Only the final decoder depends on the specified signal strength.

The peak is $n + 1$; sampling and retaining the final binary answer uses two states. Bound (12) gives the converse even with adaptive acquisition. For a fixed acquisition order, after t bits fix all future input bits and consider queries among the first t coordinates. The resulting continuation rows have affine dimension t , forcing $t + 1$ states at that cut. This proves simultaneous minimality of the claimed streaming profile. $\square$

4.3. A geometry stronger than rank.

Proof of (10). For a corner x and a possible generator $z \in [-1, 1]^n$, set

$$d_x(z) = \frac{1}{2} \sum_{i=1}^n (1 - x_i z_i).$$

Equation (11) implies that its average under the encoder for x is $n(1 - \eta)/2 < 1/2$. Hence some used generator has $d_x(z) < 1/2$. For distinct corners x, x' , at a coordinate where they differ the two corresponding summands add to one; all other summands are nonnegative. Thus $d_x(z) + d_{x'}(z) \geq 1$, and no generator can serve both corners. All 2^n corners need distinct generators. Retaining the entire input word and then applying the single noisy query row gives the upper bound 2^n . The factorization argument is independent of the acquisition order or its internal randomization. $\square$

For $n \geq 2$, both exact regimes contain full-support laws. In the strong regime the ordinary rank is only $n + 1$, but the exact causal memory is 2^n . In the weak regime the nested simplices attain that rank with a compatible streaming implementation. Thus neither support alone nor linear dimension alone determines exact memory. In persistent bits these two regimes give respectively $\lceil \log_2(n + 1) \rceil$ and n . At $\eta = 0$ the query-state complexity is one; the peak is two if the emitted answer must be held for a charged output cut. Every nonzero weak signal has query-state complexity $n + 1$, despite full support and uniform convergence to the fair-output channel as $\eta \downarrow 0$.

Proposition 4.2 (Known rows, approximation, and a strict rank margin). *The input-independent fair-output channel has rowwise total-variation error $\eta/2$ in approximating (8). For $n \geq 2$, any approximation factoring through at most n query states has worst rowwise error at least $\eta/(2\sqrt{n})$. The exact state counts in Theorem 4.1 refer to known η and atomic stochastic rows, not to an unknown calibration or a fixed-length finite fair-bit implementation.*

Proof. The first assertion is the binary total-variation formula. For the second, form the 2^n by $2n$ probability matrix A . Its nonzero singular values are $\sqrt{n2^{n-1}}$ and $\eta\sqrt{2^{n-1}}$ with multiplicity n . This follows by using the orthogonal row functions $1, x_1, \dots, x_n$ and, in each query block, the orthogonal column vectors $(1, 1)$ and $(1, -1)$. A matrix of rank at most n differs from A

in Frobenius norm by at least its smallest nonzero singular value. If every binary-query row has total-variation error at most δ , each pair contributes at most $2\delta^2$, so the squared Frobenius error is at most $2n2^n\delta^2$. The claimed bound follows. This is only an approximation bound, not an assertion that the two sides match. $\square$

Random-access coding is a classical communication task [22, 23, 24]. Equation (8) fixes every conditional decoding probability, not merely a lower bound on worst-case success or an average over inputs and queries. The new causal assertion made here is the exact finite-horizon profile from (14), together with lower bounds that survive adaptive acquisition and all persistent-randomness accounting. No priority claim for the communication model or for convex coding is made.

5. FROM SADDLE CONTACT TO INTRINSIC MEMORY

The geometric theorems classify response laws. A statistical optimization must first determine which laws are forced. We make that logical order explicit, so that Bayes indifference is not mistaken for freedom to choose the cheapest response.

Let $\mathcal{C}$ be the finite polytope of full-history causal response arrays and $g_\theta(C)$ the continuous payoff against a fixed unknown parameter in a compact space. Suppose μ_* is a least-favorable prior with value U . The optimal set is

$$\mathcal{S}_* = \left\{ C : \int g_\theta(C) d\mu_* = U \right\} \cap \bigcap_{\theta} \{ C : g_\theta(C) \geq U \}.$$

For every $C \in \mathcal{S}_*$, nonnegativity and zero prior integral give $g_\theta(C) = U$ on $\text{supp } \mu_*$. Differentiability at an interior supporting parameter also gives $\nabla_\theta g_\theta(C) = 0$. These are the contact equations proved in Theorem A.7.3; they apply even when a response coordinate has zero Bayes coefficient.

Corollary 5.1 (A contact-rigid saddle classification). *Suppose the saddle conditions determine a unique complete causal response array C_* . If its normalized residual geometry is componentwise simplicial at every charged cut, the minimum profile attaining U is exactly $(s_t(C_*))_t$, simultaneously. Every smaller profile misses U by a strictly positive amount in a fixed finite interface. More generally, any finite family of common normalized continuation rows forced for every $C \in \mathcal{S}_*$ gives the support-component rank lower bound at its cut, even when those rows are obtained by parameter-independent averaging.*

Proof. Any value-attaining array belongs to $\mathcal{S}_*$, and hence equals C_* in the first assertion. Apply Theorem 2.2. The normalized finite-row realization space for a smaller fixed profile is compact. If its maximal worst-case score were U , it would contain an implementation of C_* , contrary to the lower bound. For the last assertion, each forced row factors through the same state continuations at that cut. Averaging encoder rows preserves normalization and this factorization. The separated-component lower argument therefore applies to the forced rows. This last lower bound does not, without completeness and compatibility, assert an exact whole-schedule profile. $\square$

This corollary states a checkable implication after contact has been solved; it does not assume that every saddle is response-rigid. The following application deliberately uses only the common forced rows, because the marked saddle does not determine every coupling of fresh validation coins.

5.1. The marked decision geometry as a complete instance. The full-support saddle of Theorem A.15.1, proved in the supporting development, concerns

$$(\gamma, \kappa) \in [6/25, 13/50] \times [0, 1/1000].$$

Its contact equations force the following five averaged response rows on the diagonal atoms 000, 001, 110, 111, with zero response at the four other atoms:

$$(16) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t), \quad 0 < t < 1.$$

The inner three rows are individually forced by strict Bayes signs. The outer rows are averages over the independent training marks; contact and stationarity fix their formerly indifferent coordinates. Every actual state encoder still factors these averages through its same decoder. These facts are proved without a count-only restriction on competitors in Theorems A.14.1 and A.15.1.

Proposition 5.2 (Canonical five-state decision capacity). *For the forced response rows (16), the support graph has components of sizes 2, 1, 2, with affine dimensions 1, 0, 1. Their hulls are simplices. Thus their intrinsic capacity is*

$$(17) \quad 2 + 1 + 2 = 5.$$

This recovers the exact decision count against all the original stochastic encoders, training gates, stopping rules, and independent calibration, not just against a chosen five-event compiler.

Proof. Convert each response coordinate $h(y)$ into the two probabilities $(1 - h(y), h(y))$. This affine embedding identifies the response cube with its normalized binary-query channel polytope. The first two rows have intersecting minimal faces: the edge of the first contains the second. The last two rows have the reflected relation. Their faces are disjoint from one another and from the singleton face of the middle row. Hence these are exactly the three support components. The first and last components each contain two distinct points; the middle contains one. Corollary 5.1 gives the five-state lower bound. The retained optimal event selection attains it and its validation implementation is proved in the supporting physical development. No new physical hypothesis is inserted. $\square$

For m independent copies of this specified binary-query continuation, the exact joint checkpoint capacity is 5^m by Theorem 3.1. This requires the product output law for the independent copies. It is not a claim that the minimax value of an m -block audit is the product of U_2 , or that marginal optimality alone forces independent output randomization.

5.2. Known-target physical exactness and acquisition. The supporting proofs retain the earlier rational certificate and uniform two-preparation construction, with constants

$$(18) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

These are not substituted for the later exact saddle value U_2 .

For every specified Brownian target $Q_{d,\sigma}$ in $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the exact target-identification proof gives

$$\gamma = \frac{1}{4} - \frac{1}{2}\mathbb{E}e, \quad \kappa = \frac{1}{2}\mathbb{E}[e(1 - e)] > 0.$$

The conditional sign-error rate e is shared through the physical signal; replacing $\mathbb{E}e^2$ by $(\mathbb{E}e)^2$ is not justified. The retained full-support saddle and schedule theorems therefore give five decision states and minimum peak twelve over all externally declared, order-preserving validation clocks. Exactly the serial orders attain that minimum. The ideal reverse-order support limit has peak

ten. These are theorems about the specified target with its exact stochastic row available, and their proofs remain in full below.

An acquired target is different. Its finite counter and finite-bit construction in Proposition A.16.3 gives approximate score recovery with its stated calibration error and workspace. Neither that proposition nor Theorem 4.1 identifies an unknown algebraic row exactly from finitely many samples. The new intrinsic classifier does not change this distinction. Nor does its declared-path corollary convert the physical fixed-clock optimum into a lower bound for all report-dependent validation schedulers. The adaptive-acquisition quantifier of Theorem 4.1 is proved for that experiment by the barrier factorization, not imported from the collision example.

6. THEOREM-LEVEL RELATION TO REALIZATION AND CODING

There are three different minimization problems in the argument, and their classical precedents should not be conflated.

Positive realization versus an intrinsic complete class. Heller’s positive-realization viewpoint and the HMM realization survey of Vidyasagar [17, 18] concern the representation of stochastic laws by positive state systems. They motivate the normal form used here. Our main theorem does not replace that theory or solve its unrestricted minimization problem. Its finite input is a complete causal response array with declared cuts. From all its normalized residuals, it computes a face-intersection graph and a sum of affine ranks. Componentwise simpliciality is a finite geometric test, and under that test the lower sum equals a simultaneously realizable profile. No choice of a witness family remains in the answer. The general normal-form statement alone does not supply this all-realizations lower bound or its completeness test.

Residual minimality versus all-machine minimality. Denis and Esposito [21], Proposition 16 and their probabilistic residual-automaton results, establish canonical minimal generating subsets of residuals under their finite-generation hypotheses. The residual-vertex upper construction in our proof belongs to that line of ideas; its existence is not claimed as a new principle. But being smallest among residual-state representations need not mean being smallest among all positive representations. Our lower proof allows arbitrary state continuations outside the residual hull and rules out their sharing across distinct support components. It is this additional step, together with affine independence inside each component, that equates residual size with unrestricted stochastic size in the stated class. The square example after Corollary 2.4 exhibits failure of rank completeness outside that class.

Nested polytopes versus causally compatible covers. Gillis and Glineur [19], Definition 1 and Theorem 1, distinguish restricted nonnegative rank and give its nested-polytope formulation. Their distinction between ordinary and restricted rank is retained: we never force a competing generator into the residual affine span. For normalized binary-query channels the cube containment in (11) follows directly by conditioning. That checkpoint characterization is not a new identification of nonnegative factorization with convex geometry. The additional online question is whether covers at successive cuts admit stochastic shifts. The explicit inclusions (14) answer it for every preparation length and attain the lower rank at every acquisition cut. A final enclosing simplex without those inclusions would not establish Theorem 4.1.

Reusch and Merzenich [20] study compatible coverings for incompletely specified sequential machines. Their subject is particularly relevant when a target has support zeros. We do not infer a canonical completion from an almost-sure law. The present intrinsic theorem uses a complete array, and the retained physical support-boundary proof specifies its completion and supported tests explicitly.

Random-access codes versus exact online channel synthesis. Classical and quantum random-access coding, including the distinction between private and shared randomness, is established

in [22]. Recent work studies average and worst-case coding probabilities [23, 24]. The classical private-randomness factorization in [24], equation (4), is precisely the checkpoint conditioning mechanism underlying (11). Thus the random-access model and that factorization are not inventions of this article. The present statement fixes every conditional probability to (8) and prices the entire sequential encoder. The affine-dimension lower bound, the compatible $(t + 1)$ -state streaming construction, and the disjoint corner-cap converse give two exact regimes with different memory scales in one growing full-support family. We do not claim the optimal success probability for every message size or a complete intermediate-signal phase diagram. Statistical contact and the scope of originality. The new geometric classification applies to laws already specified or forced. A minimax problem supplies those laws only after Bayes optimality and uniform contact have both been imposed. The marked application makes this distinction concrete: the cheaper three-event Bayes tie rule is not uniformly minimax. Contact forces the two extra coordinates before intrinsic realization is considered. This comparison isolates the hypotheses and conclusions being used; it is not a claim that no reformulation of earlier theory could imply them. No complete priority or proof-level non-overlap audit of Norberg's filtered-experiment comparison [11] is asserted. The finite geometric and online results above are proved directly, with the inherited comparison, control, and testing mechanisms attributed in the supporting text.

SUPPORTING FOUNDATIONS AND PHYSICAL PROOFS

The following sections preserve the substantive mathematical body of revision 28. They supply the exact saddles, microscopic model, validation-order proofs, general causal foundations, and quantitative consumers used above. Earlier local scope statements describe the individual results in which they occur; the new classification and streaming theorems are stated in Sections 2–5. All predecessor theorem labels and all mathematical proofs remain available in this manuscript. The current introduction and theorem hierarchy replace repeated roadmap prose, not the supporting mathematics.

APPENDIX A.1. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

A.1.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any

justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition A.1.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t | h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t | x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

A.1.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem A.1.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y}f)}{\pi(\mathbb{T}_{t,a,y}1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition A.1.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$D(P\|Q) = D(PG\|QG) + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw).$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

APPENDIX A.2. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

A.2.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition A.2.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(19) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed, each with risk at least (19). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem A.2.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z, a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for*

all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies

$$(20) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition A.2.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most Le_t . Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

A.2.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition A.2.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem A.2.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(21) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(22) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (21). More explicitly, with $D_0 = d_0 + a_0 a + \delta_0$ and $D_j = d_j + a_j a + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (22). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition A.2.3. $\square$

Corollary A.2.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0 d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem A.2.4 can use $N_{\text{cal}} = r_0 n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most $r_0 d$ coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section A.6 is automatically an executable finite register.

APPENDIX A.3. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition A.3.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(23) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition A.3.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (23) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row.

Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section A.21 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

APPENDIX A.4. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(24) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z)(Q - P_\theta)h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem A.4.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(25) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(26) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (26) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (24) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (25). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (26). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem A.4.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (24). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (24). $\square$

Remark A.4.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (26) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary A.4.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (25) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (25) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

APPENDIX A.5. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh

independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(27) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y)d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(28) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (27) is the mass times the largest posterior-predictive total variation.

Theorem A.5.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(29) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (27), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (28) and identifies the right side with (29). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (29) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary A.5.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (28) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition A.5.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

A.5.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition A.5.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(30) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	1/2	5/8	5/8
L	5/8	7/16	1/2
H	5/8	1/2	7/16

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by 5/8; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is 3/4. Otherwise only i is known, and the complement of its two-atom fiber gives score 1/2. Each fixed parameter therefore has mean score 5/8.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition A.5.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

APPENDIX A.6. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

A.6.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(31) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(32) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem A.6.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(33) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (32) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H in (33) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (33) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

A.6.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s | i)$. It then chooses one of the m sensing actions with probabilities $D(a | s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem A.6.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(34) \quad V_{m,K} = \frac{1}{2\lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(35) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (34) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s | i) D(i | s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i | s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i | s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s | i) D(i | s) \leq \frac{1}{m} \sum_{i,s} D(i | s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary

singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (35). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (34) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary A.6.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem A.6.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (32).

APPENDIX A.7. SADDLE FACES AND CAUSAL REALIZATION

The statistical saddle and its implementation are different objects. The former is convex; the latter need not be. The purpose of this section is to retain that distinction while extracting memory obstructions from an unrestricted saddle. In particular, a zero Bayes coefficient must not be discarded before the contact conditions at the worst parameters have been imposed.

A.7.1. A normal form over a finite causal interface. Fix a finite horizon and finite action and report alphabets. An action a_t is chosen before report y_t . A terminal output is included as one last action, followed by a dummy report. Illegal actions can be removed from each protocol type; equivalently, use a fixed padding of the finite protocol tree and impose its zero constraints. Define a *causal response array* $C(\mathbf{a} \mid \mathbf{y})$ by nonnegativity, normalization in $\mathbf{a}$ for every $\mathbf{y}$, and the requirement that its marginal on $a_1, \dots, a_t$ depends only on $y_1, \dots, y_{t-1}$. The resulting set $\mathcal{C}$ is a finite polytope. Conditioning successive marginals realizes every such array by a full-history randomized controller; arbitrary versions on zero marginals do not change the array.

For a fixed parameter θ , the environment has a history-dependent likelihood $L_\theta(\mathbf{a}, \mathbf{y})$, obtained by marginalizing its hidden states. The bounded reward has the linear form

$$(36) \quad g_\theta(C) = \sum_{\mathbf{a}, \mathbf{y}} L_\theta(\mathbf{a}, \mathbf{y}) r_\theta(\mathbf{a}, \mathbf{y}) C(\mathbf{a} \mid \mathbf{y}).$$

Assume its coefficient vector is continuous on a compact metric parameter space. The parameter is fixed once, not selected after a report. Protocol-specific equalities, including a frozen validation

choice, are represented by the corresponding finite protocol tree. A discarded protocol choice is not supplied again as a free input.

Here is an explicit positive realization of a response array. At cut t use k_t states, an initial distribution α , and nonnegative matrices $M_t^{a,y}$ of size k_{t-1} by k_t such that

$$(37) \quad M_t^{a,y} \mathbf{1} = p_t(a), \quad \sum_a p_t(a) = \mathbf{1}, \quad p_t(a) \text{ is independent of } y.$$

Then

$$(38) \quad C(\mathbf{a} \mid \mathbf{y}) = \alpha M_1^{a_1, y_1} \cdots M_T^{a_T, y_T} \mathbf{1}.$$

The independence in (37) is essential: otherwise an action could use its not-yet-observed report. If an action or an input must survive an intermediate cut, split this formula at that cut and include its buffer in the state alphabet. Likewise, impose any declared shared-row constraints before optimizing. An externally fixed clock permits time-dependent matrices; an autonomous implementation has to store its phase or use the prescribed shared rows.

Proposition A.7.1 (Outer normal form and compatible continuations). *For the finite interface and declared cuts, the response arrays of all controllers with profile at most $\mathbf{K}$ are exactly the arrays (38) with $k_t \leq K_t$, the normalization (37), and the declared interface constraints. This set, denoted $\mathcal{C}_{\mathbf{K}}$, is compact. The assertion is over all wirings of the allowed register, not one chosen transition graph.*

Equivalently, there are at most K_t normalized continuation arrays $B_{t,s}$ at each cut, with terminal continuation 1, and common nonnegative coefficients satisfying

$$(39) \quad B_{t-1,s}(a\mathbf{a} \mid y\mathbf{y}) = \sum_{s'} M_t^{a,y}(s, s') B_{t,s'}(\mathbf{a} \mid \mathbf{y}), \quad C = \sum_s \alpha(s) B_{0,s}.$$

Separate positive factorizations at the cuts are not sufficient without these common shift identities.

Proof. In a controller, let $p_t(a \mid s)$ be its action row and $q_t(s' \mid s, a, y)$ its update row. Their product defines $M_t^{a,y}$ and gives (37). Multiplication over a complete execution gives (38). Every finite wiring embeds into the fully connected register by assigning zero to unused transitions; smaller alphabets are padded by unused states. Thus no further maximization over graph presentations is necessary.

Conversely choose action a with probability $p_t(a \mid s)$, and, after observing y , use the row $M_t^{a,y}(s, s')/p_t(a \mid s)$ when the denominator is positive. At a zero denominator the entire corresponding nonnegative row vanishes; choose an arbitrary legal update there. This gives exactly the required runtime arguments. It does not expose a posterior, the parameter, or an offline occupation array to the machine. Apply the same construction at each declared buffer cut. A product of finitely many stochastic row simplices is compact; closed legality and sharing constraints preserve compactness. Its continuous image is $\mathcal{C}_{\mathbf{K}}$.

The future execution starting at a state s defines $B_{t,s}$ and satisfies (39). Conversely, induction from the terminal continuation and these identities gives the product formula. Thus the continuation description has exactly the same realizations. In particular, independently selected factorizations at different cuts cannot be spliced unless the shift equations hold. $\square$

This is a finite controlled positive-realization statement. The role of positivity and invariant generating cones in stochastic realization is classical [17, 18, 19]. Ordinary linear rank need not be a state count. Here the additional statistical question is which positive realizations meet an optimal saddle face. Proposition A.7.1 provides a normal form, not an efficient factorization algorithm. It also does not compactify an unbounded number of observable operations.

A.7.2. The realization theorem. Let

$$U = \max_{C \in \mathcal{C}} \min_{\theta} g_{\theta}(C), \quad V_{\mathbf{K}} = \max_{C \in \mathcal{C}_{\mathbf{K}}} \min_{\theta} g_{\theta}(C).$$

By compact convex minimax there is a least-favorable prior μ_* with $U = \max_{C \in \mathcal{C}} \int g_{\theta}(C) d\mu_*$. Put

$$\begin{aligned} \ell_*(C) &= \int g_{\theta}(C) d\mu_*, \\ F_* &= \{C \in \mathcal{C} : \ell_*(C) = U\}, \\ \mathcal{S}_* &= F_* \cap \bigcap_{\theta} \{C : g_{\theta}(C) \geq U\}. \end{aligned} \tag{40}$$

The set F_* is an exposed face. The second intersection is important: a Bayes best response to a least-favorable prior need not be minimax.

Here $\mathcal{C}$ is the full-history convex reference for the finite protocol. A nonconvex restriction, such as reuse of one stochastic table at several visits, is imposed on $\mathcal{C}_{\mathbf{K}}$, not inserted into $\mathcal{C}$ before invoking convex minimax. In that case U is a reference upper value unless exactness for the unrestricted physical class is separately known. In the reset applications below that exactness is supplied by Theorems A.4.1 and A.4.2.

Proposition A.7.2 (Bayes loss and worst-case loss). *Under the preceding assumptions,*

$$U - V_{\mathbf{K}} = \min_{C \in \mathcal{C}_{\mathbf{K}}} \left\{ U - \ell_*(C) + \ell_*(C) - \min_{\theta} g_{\theta}(C) \right\}. \tag{41}$$

Both summands are nonnegative and the minimum is attained.

Proof. The prior average is at most U and at least the minimum over fixed parameters. The sum in braces equals $U - \min_{\theta} g_{\theta}(C)$. Minimizing on the compact realization set proves the assertion. $\square$

Theorem A.7.3 (Saddle-face contact and compatible realization). *Under the preceding assumptions, $V_{\mathbf{K}} = U$ if and only if some array in $\mathcal{S}_*$ has compatible positive continuation generators of profile at most $\mathbf{K}$. If no such generators exist, $V_{\mathbf{K}} < U$ strictly.*

For $C \in \mathcal{S}_$ one has $g_{\theta}(C) = U$ on $\text{supp } \mu_*$. At an interior supporting parameter at which the payoff is differentiable, $\nabla_{\theta} g_{\theta}(C) = 0$. These equalities and stationarity conditions hold also in directions whose Bayes coefficient vanishes.*

At any reset continuation cut, let R_h denote the conditional channel of future outputs under each externally specified future input word. It factors as

$$R_h = \sum_{s=1}^{K_t} E(s \mid h) B_s, \quad E(\cdot \mid h) \in \Delta_{K_t}, \tag{42}$$

with one common family of continuation channels B_s . Any obstruction to this positive factorization for every member of $\mathcal{S}_$ is therefore a memory obstruction for every admissible architecture, regardless of its training wiring.*

Proof. By Proposition A.7.2, the two nonnegative losses vanish exactly when C belongs to $\mathcal{S}_*$. Proposition A.7.1 then proves the realization criterion. Compactness makes nonintersection a strictly positive value loss, not merely failure of one optimizer.

For an optimal array, $g_{\theta}(C) - U$ is continuous, nonnegative, and has zero μ_* integral. It is zero on the support of that measure. Every interior zero is a local minimum, so its gradient vanishes. No strict-sign hypothesis on individual coordinates is needed. Finally condition on the actual retained state at a reset cut. Given that state, the future program receives the same

specified inputs and has the same output law, irrespective of the discarded past. Averaging gives (42). All persistent random variables are part of that state. The assertion can equivalently be checked on controller input tapes, using the generators in (39); it does not assume that a hidden physical state resets at a cut where the interface has not reset it. $\square$

The identity (41) is an exact loss decomposition, not a new interchange of a minimum and maximum over nonconvex machines. Its use is that an unrestricted certificate identifies a finite face, while contact at its supporting parameters can determine the remaining coordinates. The resulting channels are then tested for causal realizability. This is stronger than choosing an arbitrary Bayes tie rule and counting the states of that one rule.

Corollary A.7.4 (Explicit coefficient form for frozen validation). *In the reset experiment of (24), write*

$$C_{z\omega}^* = \int P_{\theta}^{\otimes N}(z)(Q(\omega) - P_{\theta}(\omega)) d\mu_*(\theta).$$

The first summand of (41) is exactly

$$(43) \quad \sum_{C_{z\omega}^* > 0} C_{z\omega}^*(1 - h_z(\omega)) + \sum_{C_{z\omega}^* < 0} |C_{z\omega}^*| h_z(\omega).$$

Thus optimality fixes every strict-sign coordinate, while the support contact and stationarity equations must still be imposed at the zeros. For K labels at the post-training decision cut, an arbitrary upstream encoder gives the compact relaxation

$$(44) \quad h_z(\omega) = \sum_{s=1}^K E(s | z) D_s(\omega), \quad E \in (\Delta_K)^{\Omega^N}, \quad D_s \in [0, 1]^{\Omega}.$$

If this relaxation misses the saddle set, its strictly positive gap bounds every smaller architectural class.

Proof. Maximizing each coefficient separately gives the sum of its positive part. Subtraction gives (43). Conditioning on the decision state gives (44); the post-cut randomization is independent of the discarded word. Conversely an arbitrary encoder and a response list realize this factorization when other memory is unrestricted. The parameter spaces in (44) are compact, so the final assertion follows from Theorem A.7.3. $\square$

A.7.3. Faces as memory witnesses. For a finite input set $\mathcal{Y}$ and output alphabet $\mathcal{D}$, continuation channels form the polytope $\mathcal{P} = \prod_{y \in \mathcal{Y}} \Delta(\mathcal{D})$. Its vertices are deterministic output functions. The analogous polytope for scalar test responses is $[0, 1]^{\mathcal{Y}}$.

Lemma A.7.5 (Vertex and face packing). *Suppose channels to be factored through one common state alphabet include s distinct vertices $v_1, \dots, v_s$ of a polytope $\mathcal{P}$ and rows $r_1, \dots, r_j$. Suppose $r_i \in F_i$, the faces F_i are pairwise disjoint, and*

$$r_i \notin \text{conv}\{v_a : v_a \in F_i\}.$$

Every factorization of these rows as stochastic mixtures of generators in $\mathcal{P}$ needs at least $s + j$ generators.

Proof. A convex decomposition of a vertex uses that same vertex at every positive coefficient. Hence all s vertices must occur among the generators. If a convex combination belongs to a face, all generators with positive coefficients belong to that face. The stated exclusion therefore forces an additional generator in each F_i . Pairwise disjointness prevents any such generator from serving two faces. $\square$

The lemma does not assert that a stochastic state must itself be a history residual. Its generator can be any channel in the ambient polytope. This is why the argument also excludes randomized encoders and randomized continuations. It is a positive-factorization argument, not ordinary matrix rank or a restriction to a deterministic compiler.

Proposition A.7.6 (Resource equivalence and stable obstructions). *A causal, cut-preserving recoding with a causal inverse, both using only one transducer state and preserving the payoff, preserves every $V_{\mathbf{K}}$, U , and the set of profiles attaining U . More generally suppose simulations in both directions have defect at most ϵ and require at most R_t states at cut t . For rewards in $[-1, 1]$,*

$$V_{\epsilon}(\mathbf{RK}) \geq V_{\mathfrak{F}}(\mathbf{K}) - 2\epsilon, \quad V_{\mathfrak{F}}(\mathbf{RK}) \geq V_{\epsilon}(\mathbf{K}) - 2\epsilon.$$

For a common controller class and training-validation kernels whose values differ by at most η , a gap $U - V_{\mathbf{K}} = \delta$ remains at least $\delta - 2\eta$ after the perturbation.

Proof. Transport a controller through each simulator and retain the product of its state and the simulator state. Apply the uniform bounded-reward variation bound and optimize. With an exact one-state inverse the two inequalities are equalities at the same profile. In the last assertion both the unrestricted and the constrained optimum change by at most η ; subtract their two bounds. The state products must be applied at the actual buffer cuts, not only at the terminal decision. $\square$

These statements place the saddle face, the future-test quotient, and resource-aware causal morphisms in one theorem chain. The channel factorizations are presentation-independent under the stated causal recodings. The memory obstruction in the next section is derived from the experimental coefficients themselves, not from a metric dimension or a covering exponent supplied as an additional hypothesis.

APPENDIX A.8. FACIAL CERTIFICATES FOR SCHEDULE-OPTIMIZED MEMORY

A schedule is part of the resource problem. The following criterion extracts a whole family of schedule converses from a saddle, rather than minimizing the states of one selected program. Its hypotheses are finite identities of continuation functions and faces. Its conclusion applies to arbitrary stochastic state encoders.

Consider finitely many validation tapes, with a fixed internal order on each tape and finitely many symbols per slot. A *declared-clock schedule* is a word specifying which tape supplies each successive slot; it is fixed before training and is independent of all reports. Let $\mathcal{G}$ be the finite acyclic graph of consumed-prefix counts. Its directed paths from the initial vertex to the final vertex are the legal schedules. This includes all interleavings, not only serial ones. The lower bound below does not give an adaptive scheduler free storage: a report-dependent choice of the next tape belongs to a different controlled interface and requires its own charged rows.

At a vertex v of $\mathcal{G}$, a retained state supplies a conditional channel on the unread tape suffixes. Assume in this section that every prefix and suffix combination used in a witness has positive probability under a supporting experiment. One may instead restrict to a stated common suffix subalphabet with this property. The channels form a product of probability simplices, denoted by $\mathcal{P}_v$.

Definition A.8.1 (Contact-forced facial certificate). A certificate at v consists of distinct deterministic channels $d_1, \dots, d_s$ and channels $r_1, \dots, r_j$, such that every minimax-optimal auditor has these rows among its conditional continuations, allowing parameter-independent averaging of specified past rows. Each r_i lies in a face F_i of $\mathcal{P}_v$, the faces are pairwise disjoint, and

$$r_i \notin \text{conv}\{d_a : d_a \in F_i\}.$$

Its certified capacity is $s + j$. Define $c(v)$ to be the largest such capacity within a declared finite witness family. Conditional averaging must retain the same future-input coordinates and may not add the unknown parameter as an input. A channel of mean zero is not thereby a deterministic zero channel.

For an upper certificate choose a minimax-optimal finite randomization over frozen deterministic events, with the selected event retained at the decision cut. Let $\mathcal{D}_v$ be its distinct future deterministic output functions after all possible supported prefixes at v . When the input support is a full product, equality of these functions is closed under every next-symbol restriction. Put $r(v) = |\mathcal{D}_v|$. More general supported experiments may use any explicitly supplied shift-compatible deterministic completion; merely minimizing separate partial-function covers at different cuts is insufficient.

Theorem A.8.2 (Facial capacity and schedule realization). *Suppose contact with the unrestricted saddle supplies the preceding certificates. For each declared-clock path π and every stochastic auditor on that path attaining the unrestricted value,*

$$(45) \quad W(\pi) \geq \max_{v \in \pi} c(v).$$

A shift-compatible deterministic residual family realizes the selected optimal event rule simultaneously with widths $r(v)$ along any such path. If its training and selection peak is W_0 , then

$$(46) \quad \min_{\pi} \max_{v \in \pi} c(v) \leq W_{\text{clk}}^{\min} \leq \min_{\pi} \max \{W_0, \max_{v \in \pi} r(v)\}.$$

In particular, a certificate and a realizing path with the same peak identify the exact minimum over all declared-clock schedules and all stochastic architectures on those schedules. If $c(v) = r(v)$ at every vertex and W_0 does not exceed the path peak, the entire minimum-profile problem reduces to the intrinsic capacity field on $\mathcal{G}$. The peak optimum of that field is computed by

$$(47) \quad b(v_0) = c(v_0), \quad b(v) = \max\{c(v), \min_{u \rightarrow v} b(u)\}.$$

The result is invariant under bijective, causal, slot-preserving recodings of the tape alphabets.

Proof. Condition on a charged retained state at v . Every past row is a stochastic mixture of the same at most $W(\pi)$ future channels. Parameter-independent averaging preserves this factorization. The face-packing lemma therefore gives at least $s + j$ states for each certificate, proving (45). This step ranges over all state wirings and all stochastic encoders, not a specified graph.

For the upper bound retain the current residual output function. If two residuals agree on the future input product, their restrictions after each legal next symbol agree as well. Therefore the next retained label is determined by the old label and that symbol. At the last cut it is the output itself. This constructs all cuts compatibly in one machine. The initially sampled event and its training selector are charged in W_0 . Optimizing the two bounds over the finite path set proves (46). The recursion follows by classifying paths according to their last edge and taking the maximum with the cost of the final vertex. Induction proves it on the acyclic graph. Finally, a bijective recoding induces an affine isomorphism of every channel polytope, preserving vertices, faces, conditional averages and restriction maps. It preserves both certificates and realizing paths. $\square$

The general positive-realization normal form and residual quotient have classical antecedents [17, 18, 19]. The minimization of partially specified machines likewise requires compatible coverings, not unrelated cutwise minima [20]. The contribution used here is different: the witness rows are forced by *every member of the uniform saddle*, through contact on its Bayes-indifferent coordinates. This makes the lower field independent of a chosen optimal program.

The criterion is a certificate theorem for this class of experiments, not an assertion that arbitrary nonnegative-rank problems have become easy or that facial packing is always exact.

APPENDIX A.9. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (24). Set

$$(48) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem A.9.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(49) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(50) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(51) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(52) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(53) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(54) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (50) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (52). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both

are nonnegative. Equality holds at both points of (51), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (51) in Theorem A.4.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(55) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (52); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (54). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (49), optimality of the response, and least favorability of the prior. $\square$

Corollary A.9.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(56) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem A.9.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

A.9.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (55) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (54). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit

prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition A.9.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem A.4.1 with $N = 0$ gives the matching upper bound. $\square$

APPENDIX A.10. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition A.10.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

A.10.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare

$(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(57) \quad \begin{aligned} r &= (-a_0, (2B-1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W-1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma A.10.2 (Uniform collision signal). *For every preparation in (57) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B-1)v_{1,y} > 1/2, \quad (2B-1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B-1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B-1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

A.10.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem A.10.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

(58)
$$\text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition A.10.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma A.10.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (58) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

APPENDIX A.11. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1-p)(1-q)$ and $c = pq$. On the ordered atoms

$$(59) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(60) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (60). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition A.10.1 and Theorem A.10.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem A.11.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (18). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(61) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

A.11.1. An exact rational certificate against one preparation.

Lemma A.11.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem A.4.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (25) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (59), the signed coefficients in (25), after grouping by K , are the entries of

$$(62) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1 - p_i)^{2-k} \left(Q_{*,j} - \frac{(1 - p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (62) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (62) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$.

A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

A.11.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (59) and with indicator zero at every off-diagonal atom:

$$(63) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma A.11.3 (Exact uniform score of the selector). *Against the ideal target, the selector (63) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (63) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(64) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

A.11.3. Causal compilation and the full register profile.

Lemma A.11.4. *The selector and its two validations admit the deterministic profile (61). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (63). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem A.11.1. Lemma A.11.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas A.11.3 and A.11.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary A.11.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem A.11.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem A.10.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark A.11.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem A.12.1 and Corollary A.12.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width over all validation orders and the full boundary for $3 \leq W \leq 11$ remain separate questions. Theorem A.14.3 below determines the peak for forward serial validation, and

Theorem A.14.1 determines the order-independent decision cut. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary A.11.7 (The enlarged physical preparation segment). *With the full architecture of Definition A.10.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition A.10.1. Theorem A.4.2 and Corollary A.9.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas A.11.3 and A.11.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

APPENDIX A.12. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section A.11. Let r be the unique root in $(1/3, 7/20)$ of

$$(65) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(66) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (65) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem A.12.1 (Exact two-preparation value). *For the full marked-word response class of (24),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass 1/2 to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(67) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (65)–(66), not by these decimals.

Proof. Write $a = (1-p)(1-q)$, $c = pq$, $u = (p+q-1)^2$ and $v = (p-q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (64). Only counts zero and four are changed by (67). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(68) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (65) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(69) \quad G(u, 0) - v_2 = \frac{(u-r)^2\{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u-v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1-p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r-1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r-1)^2(2r-3, 2r-1, 2r-1, 2r-3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (67) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem A.4.1, now gives the matching upper bound for the full response class. $\square$

Corollary A.12.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition A.10.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(70) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (63). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (67); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma A.11.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem A.12.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem A.4.2 followed by Theorem A.12.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (70). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition A.12.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

APPENDIX A.13. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section A.9 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem A.13.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(71) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(72) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (67), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem A.12.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (64) and (68). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(73) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (73) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (72).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0 : & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1 : & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2 : & -12(r-1)^2(r-1-2\gamma, r-1+2\gamma, r-1+2\gamma, r-1-2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (72). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

APPENDIX A.14. EXACT MEMORY FROM A MARKED SADDLE

The realization criterion is now applied to a saddle with genuine Bayes indifference. The zero coefficients in the two-preparation problem are not harmless: their values are fixed by contact with the parameter family. This supplies a converse for all optimal audits, rather than for a preselected response table.

Put $I = [6/25, 13/50]$. Throughout this section $r = r_\gamma$, $t = t_\gamma$, and $v = U_2(\gamma)$ are those of Theorem A.13.1. Denote the five deterministic events in (63) by $e_0, \dots, e_4$. Responses displayed below are ordered on 000, 001, 110, 111 and vanish at the other four atoms.

Theorem A.14.1 (Contact rigidity and exact decision memory). *For every $\gamma \in I$, every minimax optimal full-word response with two training preparations has, after averaging the independent training marks at the two extreme report counts, the five rows*

$$(74) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t).$$

At the three interior counts the displayed response is forced at each individual training word, not merely after averaging.

In the original reset row-event interface, the least number of states at the post-training cut that permits the exact ideal score v is five. This converse allows arbitrary upstream finite memory, randomized encoding, adaptive training gates, stopping, independent target-only calibration, and all legal post-cut randomizations. Other cuts are unrestricted in this decision-memory assertion.

Proof. First consider a response to the full two-preparation word. Under the least-favorable prior of Theorem A.13.1, every training word has positive probability. The strict coefficient signs force e_1, e_2, e_3 at report counts 1, 2, 3. At count zero only the 000 coordinate is free; at count

four only the 111 coordinate is free. All off-diagonal validation coefficients are strictly negative: the prior parameters are interior, each word likelihood is positive, and the target assigns zero mass to those atoms. Thus those coordinates are zero for every optimal response.

Let a and b be the averages of the two free coordinates over the four equiprobable training mark words at their respective extreme counts. On the diagonal parameter (p, p) the score is

$$(75) \quad H_\gamma((2p-1)^2, 0) + aA_\gamma(p) + bB_\gamma(p),$$

where

$$A_\gamma(p) = (1-p)^4 \left(\frac{1+\gamma}{4} - \frac{(1-p)^2}{2} \right), \quad B_\gamma(p) = p^4 \left(\frac{1+\gamma}{4} - \frac{p^2}{2} \right).$$

Here H_γ is the deterministic five-event score in the proof of Theorem A.13.1. At $p_+ = (1+\sqrt{r})/2$, $A_\gamma(p_+) + B_\gamma(p_+) = J_\gamma(r, 0) = 0$ and $A_\gamma(p_+) > 0$. The latter follows from $p_+ > 1/2$ and $(1+\gamma)/4 > 1/4$. Reflection exchanges A and B at $p_- = 1 - p_+$. Also $H_\gamma(r, 0) = v$. Optimality forces contact at both prior points, by Theorem A.7.3. Subtracting v from (75) therefore gives $(a-b)A_\gamma(p_+) = 0$, hence $a = b$.

With this common value the diagonal score is $H_\gamma(u, 0) + aJ_\gamma(u, 0)$, where

$$H_\gamma(u, 0) = \frac{-u^3 + (7-4\gamma)u^2 - 3u + 13 + 4\gamma}{32}, \quad J_\gamma(u, 0) = -\frac{f_\gamma(u)}{64}.$$

The supporting value r lies in $(0, 1)$, so the derivative at r vanishes. Since $f'_\gamma(r) = D_\gamma(r) > 8$, the derivative equation has the unique solution

$$a = b = \frac{64 \partial_u H_\gamma(r, 0)}{D_\gamma(r)} = t_\gamma.$$

This proves (74). Individual extreme mark words may still use different tie values; no unnecessary uniqueness is asserted.

Now take an arbitrary audit with K states at the decision cut. Restrict the candidate to the common-garbling subfamily in Definition A.10.1. Generate two all-on words and simulate the adaptive training and its stopping with independent coins. The retained state is a parameter-independent stochastic function of those two words. Conditional on that state, the frozen row-event continuation pulls back to a response $D_s \in [0, 1]^8$ on the all-on alphabet. Any calibration is part of the same encoding and retained state. Thus (44) holds without increasing K . If the audit attains v against the full physical class, its pullback attains v against this subfamily, since the unrestricted upper bound is v . Averaging selected rows preserves their factorization.

The three interior rows in (74) are distinct vertices of $[0, 1]^8$, so the generators must include e_1, e_2, e_3 themselves. The first row lies in the face

$$F_0 = \{d : d(001) = 0, d(110) = d(111) = 1, d = 0 \text{ off the diagonal}\}.$$

Of those three obligatory vertices only e_1 lies in F_0 ; since $t < 1$, this row is not e_1 . It needs another generator in F_0 . Similarly the last row needs another generator in

$$F_4 = \{d : d(000) = d(001) = 1, d(110) = 0, d = 0 \text{ off the diagonal}\}.$$

Only e_3 among the obligatory vertices lies there. The two faces are disjoint. Lemma A.7.5 gives $K \geq 3 + 2 = 5$. Conversely the optimal response mixes e_0, e_1 at count zero and e_4, e_3 at count four and otherwise selects the indicated event. Only its five-valued event label survives the cut. The inherited compiler realizes its subsequent validation with finite memory and attains v . This proves exactness of the decision count. $\square$

The contact argument is essential. Setting both free coordinates equal to one gives a Bayes best response with only the three events e_1, e_2, e_3 , but its derivative at r is nonzero because $t < 1$. It is not minimax. Counting the states of a Bayes best response alone would give an incorrect resource conclusion.

Corollary A.14.2 (A strict gap and its physical persistence). *There is $\delta_* > 0$, uniform for $\gamma \in I$, such that every two-preparation audit with at most four states at the decision cut has ideal worst-case score at most $U_2(\gamma) - \delta_*$. In particular, this bound holds for every original-interface audit of whole-schedule peak at most four. If only the target changes by total variation at most $\beta < \delta_*/2$, the five-state decision construction still strictly outperforms this class.*

Proof. Let $W_4(\gamma)$ be the maximum worst-case score over the relaxed factorizations (44) with $K = 4$ and two full training words. Their encoder and decoder spaces are compact and fixed, independently of γ . The score is jointly continuous in these variables, the target bias, and (p, q) . Consequently W_4 is continuous. By Theorem A.14.1 it is strictly less than $U_2(\gamma)$ for each $\gamma \in I$. The minimum of this positive continuous gap on compact I is positive. This defines δ_* without pretending to have evaluated it numerically. The domination argument in that theorem makes it an upper bound for every original-interface audit in question. Changing only the target changes any fixed audit's mean score by at most β . The difference between the two classes therefore decreases by at most 2β . $\square$

For the microscopic family at $\gamma = 1/4$, the signal is larger than $1/4$ and the standard deviation of each normalized sensor noise is $\sqrt{2}\sigma$. The Gaussian tail bound used in Theorem A.10.3 gives the sharper, vanishing estimate

$$(76) \quad \text{TV}(Q_{d,\sigma}, Q_*) \leq \frac{8\sigma}{\sqrt{\pi}} \exp\{-1/(64\sigma^2)\}, \quad 9/10 \leq d \leq 11/10.$$

Thus Corollary A.14.2 holds uniformly in d for all sufficiently small positive σ . This is an existence statement with the displayed physical modulus. The present proof does not certify that every $\sigma \leq 1/24$ lies below the unknown gap threshold.

A.14.1. The sharp serial-validation peak. A decision cut and the whole execution have different resource costs. We now determine the latter in a declared validation schedule. In the *forward serial class*, both training preparations finish before validation; the candidate validation's two report slots and mark are read before any target validation report is read. The same frozen row-event continuation is used on both preparations, as in Definition A.10.1. Training gates, stochastic state updates and frozen gate policies are unrestricted within that interface. No intermediate output can be reread as external memory.

Theorem A.14.3 (Sharp forward serial peak). *For every $\gamma \in I$, the least whole-schedule peak width attaining $U_2(\gamma)$ in the forward serial class is twelve. In particular, a peak of at most eleven in this class incurs a strictly positive loss, which can be chosen uniformly for $\gamma \in I$.*

Proof. Use the common-garbling subfamily and the two raw all-on training words as in Theorem A.14.1. Any optimal audit must finish both training preparations almost surely under its least-favorable prior. Indeed a stopped branch after a first preparation would have a response independent of the unused second word. For each first report count 0, 1, 2, two choices of the unused word force two different interior vertices from e_1, e_2, e_3 . Each has positive conditional probability. A positive stopped component cannot be a convex component of two distinct vertices. The same argument excludes stopping before the first draw. Null branches may be completed arbitrarily.

Consider the cut after the candidate's second report slot and before its mark. On a canonical raw input tape its continuation is a channel from $(w, y) \in \{0, 1\} \times \{000, 001, 110, 111\}$ to $\{-1, 0, 1\}$, where w is the unread candidate mark and y the fresh target word. All these target words have positive probability. Each retained state supplies one such continuation channel, common to every past leading to it. Hence the rows indexed by selected pasts factor through at most the peak number of generators.

For the strict interior training counts, the following eight rows are forced deterministic channels:

$$(77) \quad e_1(y) - f(w), \quad f \in \{0, 1 - w, 1\}; \quad e_2(y) - f(w), \quad f \in \{0, 1\}; \quad e_3(y) - f(w), \quad f \in \{0, w, 1\}.$$

For example e_1 has candidate residual 0 after an off-diagonal report pair, $1 - w$ after 00, and 1 after 11. The other cases follow by restriction of their masks. Each listed past has positive probability under both supporting candidate parameters. The eight functions are distinct: subtracting two with the same mask compares their distinct w functions; for different masks their difference depends nontrivially on y and cannot equal a function of w alone. They are eight distinct vertices of the product of output simplices.

There are four additional required rows. Average the extreme training marks as in Theorem A.14.1, and take the candidate raw report pair to be, respectively, off diagonal and 11 at count zero, and off diagonal and 00 at count four. Independence of a fresh raw candidate preparation from the frozen continuation preserves the averaged tie probability t . On the four target words these rows have means

$$(78) \quad (t, 0, 1, 1), \quad (t - 1, -1, 0, 0), \quad (1, 1, 0, t), \quad (0, 0, -1, t - 1),$$

for both values of w . More precisely, each integer coordinate here is a point mass at that integer, and the remaining coordinate is the Bernoulli mixture between the two neighboring integers. This stronger channel statement, not a statement about a zero mean alone, follows because the candidate indicator is fixed at 0 or 1 on the selected raw pair, while the target indicator is a bit. Fresh gate randomness is already part of this channel; it supplies no additional past-dependent argument to a retained state.

The four rows lie in four pairwise disjoint faces of the product of output simplices. These faces fix, for both w , respectively,

	000	001	110	111
$F_{0,+}$	*	0	1	1
$F_{0,-}$	*	-1	0	0
$F_{4,+}$	1	1	0	*
$F_{4,-}$	0	0	-1	*

where a number specifies a point mass and $*$ is unrestricted. Among the eight vertices (77), each face contains exactly one: e_1 , $e_1 - 1$, e_3 , and $e_3 - 1$, respectively. Since $t < 1$, none of the four required rows is that vertex. Face packing forces four more generators, for a total of twelve. This argument uses conditional future-output laws and not the continued storage of an event name after a residual has replaced it.

The optimal five-event selection followed by the inherited forward compiler has peak twelve, including its stochastic selection cut. It proves the matching upper bound. To see strictness below twelve, use the finite-protocol normal form of Proposition A.7.1 with all logical cuts bounded by eleven, including selection and frozen-continuation buffers. Unobserved internal random computations can be collapsed to common transition kernels on the same register; an unretained random duration cannot be supplied as a later input. The normalized row spaces and closed frozen-interface restrictions are compact. If the constrained value equalled U_2 , the

preceding face obstruction would apply. Continuity and compactness in γ give a uniform strictly positive gap, just as in Corollary A.14.2. $\square$

Theorem A.14.3 is a minimum-width theorem for a specified serial order, not a claim that an arbitrary validation order has the same minimum. The ideal target has support restrictions that can change future residuals when the target is read first. Reverse-order and interleaved validations require their own continuation analysis. The decision-memory converse in Theorem A.14.1 does not make this serial-order assumption. Neither statement replaces the original physical interface by an uncharged atomic validation.

APPENDIX A.15. A FULL-SUPPORT MARKED SADDLE

The support restriction of the ideal target can be removed without replacing the private simulator or changing the training experiment. This section identifies an explicit region of positive targets on which both the statistical saddle and its decision-memory number are exact. The next section shows that the original Brownian sensor belongs to this region throughout its previously stated noise range.

Put

$$\mathcal{R} = [6/25, 13/50] \times [0, 1/1000].$$

For $(\gamma, \kappa) \in \mathcal{R}$, define $Q_{\gamma, \kappa}$ on $\{0, 1\}^3$ by the masses

$$(79) \quad \frac{\omega}{Q_{\gamma, \kappa}(\omega)} \left| \begin{array}{cccc|c} 000 & 001 & 110 & 111 & \text{each other atom} \\ a & b & b & a & \kappa \end{array} \right|, \quad a = \frac{1+\gamma}{4} - \kappa, \quad b = \frac{1-\gamma}{4} - \kappa.$$

These masses sum to one. For $\kappa > 0$ every atom has positive mass. The candidate is still $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$; there is no additional candidate parameter.

Theorem A.15.1 (Full-support saddle and contact rigidity). *For every $(\gamma, \kappa) \in \mathcal{R}$, there is a unique $r = r_{\gamma, \kappa}$ in $(33/100, 7/20)$ satisfying*

$$(80) \quad f(r) = r^3 + (13 - 2\gamma + 8\kappa)r^2 + (3 - 12\gamma + 48\kappa)r - 1 - 2\gamma + 8\kappa = 0.$$

Set $D = f'(r)$ and

$$(81) \quad t = \frac{2\{-3r^2 + (14 - 8\gamma - 16\kappa)r - 3 + 48\kappa\}}{D},$$

$$(82) \quad v = \frac{-r^3 + (7 - 4\gamma - 8\kappa)r^2 + (-3 + 48\kappa)r + 13 + 4\gamma - 104\kappa}{32}.$$

Then $8 < D < 10$, $1/10 < t < 1/4$, and

$$U_2(b, Q_{\gamma, \kappa}) = v.$$

The optimal five response rows are (74), with this value of t , and zero at all off-diagonal atoms. The equal prior on $p = q = (1 \pm \sqrt{r})/2$ is least favorable. The same averaged rows are forced for every full-word minimax response. At the three interior counts the entire deterministic event is forced word by word.

The least post-training decision width attaining v is exactly five, including in the original gate interface with arbitrary stochastic training, stopping and independently acquired calibration.

Proof. On the indicated rectangle in (γ, κ, r) , direct monotonicity gives $8 < f'(r) < 10$. The polynomial decreases with γ and increases with κ . Its two endpoint evaluations are bounded by

$$f_{6/25, 1/1000}(33/100) = -81619/5000000 < 0, \quad f_{13/50, 0}(7/20) = 387/40000 > 0.$$

This proves existence and uniqueness in the stated interval. The numerator of t is $-16\gamma r - 32\kappa r + 96\kappa - 6r^2 + 28r - 6$. Bounding its negative and positive terms separately gives a lower

bound $10378/10000 > 1$ and an upper bound $19754/10000 < 2$. Together with $8 < D < 10$ these prove the asserted interval for t .

Let $u = (p + q - 1)^2$, $z = (p - q)^2$, and retain the polynomials H_γ, J_γ from Theorem A.13.1. The probabilities of the five training counts give

$$E(u, z) = \frac{(u - z)^2 - 6u + 2z + 13}{4},$$

$$L(u, z) = \frac{(u - z)^2 + 6u - 2z + 1}{8}.$$

Here E is the expected cardinality of the deterministic selected four-atom event, and L is the probability of an extreme training count. The score of the proposed response is therefore

$$G = H_{\gamma, \kappa} + tJ_{\gamma, \kappa}, \quad H_{\gamma, \kappa} = H_\gamma - \kappa E, \quad J_{\gamma, \kappa} = J_\gamma - \kappa L.$$

This is an identity in the actual training probabilities, not an estimate. On the diagonal, $J_{\gamma, \kappa}(u, 0) = -f(u)/64$ and $H_{\gamma, \kappa}(r, 0) = v$. Polynomial division, using $f(r) = 0$, gives

$$(83) \quad G(u, 0) - v = \frac{(u - r)^2 A(\gamma, \kappa, r, u)}{16D},$$

where

$$A = 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru + 12\gamma r + 6\gamma u - 51\gamma + 30$$

$$- 384\kappa^2 - 96\kappa r - 48\kappa u - 144\kappa.$$

For $0 \leq u \leq 1$ the negative terms alone give $A > 8$ throughout $\mathcal{R}$. Moreover, on $0 \leq u, z \leq 1$ the bounds in the preceding family theorem yield

$$\partial_z H_{\gamma, \kappa} \geq 5/32 - \kappa, \quad \partial_z J_{\gamma, \kappa} \geq -3/64.$$

Indeed the additional derivatives are respectively $\kappa(u - z - 1)/2$ and $\kappa(u - z + 1)/4$. Since $t < 1/4$, one has $\partial_z G > 1/8$. Thus (83) is a global lower certificate on the parameter square, with equality at exactly the two proposed prior points.

For the Bayes upper bound, group the four report bits by their total S ; the independent training marks and arrangements conditional on S are parameter independent under this prior. Put $L_r = r^2 + 6r + 1$. The first three rows of 256 times the signed coefficient table on 000, 001, 110, 111 are

$$S = 0 : \quad (0, -8\gamma L_r, 32r(r + 1) - 8\gamma L_r, 32r(r + 1)),$$

$$S = 1 : \quad 8(r - 1)(r^2 + 4r - 1 - 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$\quad r^2 + 4r - 1 + 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$\quad r^2 - 4r - 1 + 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$\quad r^2 - 4r - 1 - 2\gamma(r + 1) + 8\kappa(r + 1)),$$

$$S = 2 : \quad -12(r - 1)^2(r - 1 - 2\gamma + 8\kappa, r - 1 + 2\gamma + 8\kappa,$$

$$\quad r - 1 + 2\gamma + 8\kappa, r - 1 - 2\gamma + 8\kappa).$$

The other rows are their reversals. For example the potentially small first parenthesis in row one is at most

$$(7/20)^2 + 4(7/20) - 1 - 2(6/25)(133/100) + 8(1/1000)(27/20) < -1/10.$$

The third parenthesis is negative by the same endpoint bounds; the second is positive and the fourth negative. In row two, $r - 1 + 2\gamma + 8\kappa \leq -61/500 < 0$. Row zero's third entry is positive since $32(33/100)(133/100) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 7$. Consequently the signs are

exactly those of the previous two-preparation saddle. At each off-diagonal atom the candidate mass on both prior points equals $(1-r)/8 > 13/160$, whereas $\kappa \leq 1/1000$. Every off-diagonal coefficient is strictly negative. Coordinatewise Bayes maximization therefore selects the proposed response, including its two indifferent coordinates. Its equal score at the prior support closes the minimax match against the full response class.

For completeness the contact argument persists, not just the sign pattern. If a_0, b_0 are the averaged extreme response coordinates, their extra diagonal contributions are

$$a_0(1-p)^4\{a - (1-p)^2/2\} + b_0p^4\{a - p^2/2\}.$$

At either support point their sum with $a_0 = b_0 = 1$ is zero by (80); the first term is positive at p_+ . Contact at both supports gives $a_0 = b_0$. Interior stationarity gives $a_0 = b_0 = 64\partial_u H_{\gamma,\kappa}(r, 0)/D = t$. All strict coordinates are forced individually. Thus the three vertices and two disjoint faces used in Theorem A.14.1 are unchanged. That theorem's positive-factorization argument gives five states for every admissible stochastic encoder, while the same five-event selection attains it. The common-garbbling reduction is unchanged because the candidate interface is unchanged. $\square$

The rectangle $\mathcal{R}$ is an explicit stability region for the *saddle and its exact decision-memory number*. It is not a claim that arbitrary nonsymmetric total-variation perturbations have the same optimizer. At $\kappa = 0$ every formula reduces to the retained ideal theorem. Positive off-diagonal target masses do not destroy the strict negative Bayes coefficients at those atoms: their magnitude relative to the candidate, rather than their being zero, is what is needed.

APPENDIX A.16. EXACT STATISTICAL FORM OF THE BROWNIAN COLLISION SENSOR

The full-support theorem applies to the original microscopic target, not only to an independently constructed contamination model. The preparation symmetries supply the required form exactly.

Theorem A.16.1 (Physical symmetry and a quantitative noise region). *Use exactly the preparation boxes, elastic collision, detector and Brownian windows of Section A.10. Conditional on the preparation put*

$$Z = |h(v_{1,y}^+)|, \quad e = \Phi\left(-\frac{Z}{\sqrt{2}\sigma}\right), \quad \mu_1 = \mathbb{E}e, \quad \mu_2 = \mathbb{E}e^2.$$

For every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$ the exact all-on target law is

$$(84) \quad Q_{d,\sigma} = Q_{\gamma_{d,\sigma}, \kappa_{d,\sigma}}, \quad \gamma_{d,\sigma} = \frac{1}{4} - \frac{\mu_1}{2}, \quad \kappa_{d,\sigma} = \frac{\mu_1 - \mu_2}{2}.$$

In particular

$$\frac{1}{4} - \frac{1}{1200} < \gamma_{d,\sigma} < \frac{1}{4}, \quad 0 < \kappa_{d,\sigma} < \frac{1}{1200},$$

so the entire physical range lies inside $\mathcal{R}$, and every target atom has strictly positive probability. The physical two-preparation value in the unrestricted-memory original gate interface equals (82) at these parameters. Its exact decision-memory number is five.

Proof. The collision estimate gives $Z > 1/4$ for every prepared point. The conditional two window noises are independent, of equal variance, and the deterministic signal is constant after the collision. Thus, given (B, W, Z) , the two report bits are independent copies of B flipped with probability e .

We justify the distribution of Z rather than assume a noise symmetry. Reflection in the spatial y coordinate sends (r_y, g_y, V_y) to their negatives, interchanges the two B boxes, and preserves their uniform product laws and the elastic equations. It negates $v_{1,y}^+$ and leaves Z

unchanged. The only box that depends on W is the one for V_z ; changing this center-of-mass velocity cannot change the relative collision or the outgoing y velocity. All other box coordinates are independent of W . Consequently Z , and hence e , has the same conditional law for all four (B, W) labels. The Brownian increments are independent of these labels and of the preparation.

The prepared masses of (B, W) are $5/16$ for equality and $3/16$ otherwise. Direct averaging therefore gives

$$\begin{aligned} Q(000) &= Q(111) = 5/16 - 5\mu_1/8 + \mu_2/2, \\ Q(001) &= Q(110) = 3/16 - 3\mu_1/8 + \mu_2/2, \\ Q(010) &= Q(011) = Q(100) = Q(101) = (\mu_1 - \mu_2)/2. \end{aligned}$$

These are precisely (79) with (84). This calculation uses both the common signal and the actual mark; replacing the common signal by independent unconditional report errors would in general replace μ_2 incorrectly by μ_1^2 .

The Gaussian bound already proved in the physical bridge yields $0 < e < 1/600$ uniformly in this physical range. Thus $0 < \mu_2 < \mu_1 < 1/600$, proving the parameter bounds. Positivity follows also directly from $0 < e < 1$ and positive prepared label masses. The all-on construction works against every private candidate in Definition A.10.1; its score is the product-family score. Conversely the common-garbling private subfamily gives the architecture-uniform upper bound. Theorem A.15.1 therefore supplies both exact value and exact decision count for the original interface, not for an auxiliary sensor. $\square$

Remark A.16.2 (Known target and acquired calibration). The exact value and exact atomic stochastic realization are statements for a specified target law. Their event-selection probability depends on that target, not on the unknown candidate. When d, σ and the preparation are known, the expectations in (84) are defined integrals of that known physical law. They are not free information about the private parameter. An auditor that must acquire the target instead uses finite calibration and a precision allowance. The following explicit bound makes this separation operational.

Proposition A.16.3 (Acquisition and finite row precision). *Let the fixed physical target be in the preceding theorem. From n independent all-on target calibration words define*

$$\hat{\kappa} = \frac{1}{4n} \sum_{i=1}^n \mathbf{1}_{\{X_{1i} \neq X_{2i}\}}, \quad \hat{\gamma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{1}_{\{000,111\}} - \mathbf{1}_{\{001,110\}})(X_i).$$

Clip these separately to the two intervals of $\mathcal{R}$. With probability at least $1 - 4\exp(-na^2/2)$, both coordinate errors are at most a . Choose the optimal response for this clipped target and replace its tie probability by a dyadic number with error at most 2^{-b} . Conditional on that calibration event, its worst-case score against the true target is at least

$$(85) \quad U_2(b, Q_{d,\sigma}) - 9a - 2^{-b}.$$

The calibration can be acquired with at most $(n+1)^3$ counter states and at most $4(n+1)^3$ states including the current report buffer; retaining the calibration output during the audit gives the conservative product bound $12(n+1)^3$ at logical report cuts. A sequential fair-bit realization of the tie coin, with its origin and bit phase retained, has a finite explicit bound $12(n+1)^3 + 8(b+2)(n+1)^3$.

Proof. The off-diagonal indicator has mean 4κ and range one. The signed diagonal indicator has mean $2a - 2b = \gamma$ and range two. Hoeffding's inequality and a union bound give the displayed failure probability; clipping to an interval containing the truth cannot increase coordinate error. The two target laws have total variation at most $|\Delta\gamma|/2 + 4|\Delta\kappa| \leq 9a/2$. Every fixed trained

response changes score by at most this amount when only the target changes, and so does the unrestricted optimal value. Using an optimizer for the estimated target therefore loses at most $9a$ against the true one. Only the two extreme-count event choices are rounded. Their score contribution changes by at most the coin error, because their total probability is at most one and the change in a single indicator expectation has magnitude at most one. This proves (85).

Three nonnegative category counts, each bounded by n , compute the two statistics; they also fix the clipped estimates and the subsequent program choice. Keeping this count triple with the twelve-state audit costs at most the stated product. At an extreme count, sample the dyadic coin using b fair bits and comparison with its binary threshold, retaining the two-valued origin, the comparison flag and the phase. At most $8(b+2)$ additional states suffice per calibration choice. All other counts select deterministic events. The bound includes the coin workspace instead of identifying an algebraic row with a finite random-bit primitive. The n target calibrations and the four audit preparations are all charged. $\square$

APPENDIX A.17. OPTIMAL VALIDATION ORDERS AND A POSITIVE-NOISE MEMORY JUMP

We now optimize the order in the original two-tape, bit-level validation interface. Each tape has slots $1, 2, w$ in that order. Write X for the candidate tape and Y for the target tape. There are twenty declared-clock interleavings. A fixed clock is common to all executions; its phase is not a report-dependent side channel. Gate choices, state updates, training and frozen-event selection remain stochastic and otherwise unrestricted as before.

Let $e_0, \dots, e_4$ be the five events (63). A full raw validation word has output $e_i(Y) - e_i(X)$. For the strict training counts, contact forces $i = 1, 2, 3$ respectively. At either extreme only one raw-event coordinate is indifferent; Theorem A.15.1 fixes its mark-averaged probability to $t \in (1/10, 1/4)$.

One precaution is needed. A freshly randomized gate need not couple its two validation coins as one stored event mask would. When both raw validation words equal the indifferent atom, their difference may be random with mean zero. None of the lower certificates below uses that coordinate as a forced deterministic zero. For every additional mixed witness, the known prefix excludes at least one validation word from the indifferent atom, so at least one of the two event indicators is already deterministic. Its future output law is then fixed by the remaining indicator's marginal. This covers both stored-mask and fresh gate implementations.

Theorem A.17.1 (The minimum over all declared-clock validation orders). *For every $(\gamma, \kappa) \in \mathcal{R}$ with $\kappa > 0$, the least whole-schedule peak attaining $U_2(b, Q_{\gamma, \kappa})$, minimized over all twenty declared-clock interleavings, is twelve. The only minimizing orders are the two serial orders*

$$XXXYYY, \quad YYYXXX.$$

Every nonserial declared order requires peak at least fourteen. The lower bounds permit the full stochastic gate and event interface, not merely an all-on deterministic compiler. These conclusions apply to $Q_{d, \sigma}$ for every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$.

Proof. Every raw validation prefix has positive probability at either supporting candidate and at the target. Thus all cut witnesses may be compared on a common full future alphabet. Generate all-on words to simulate training gates as in Theorem A.14.1. The post-training response is an unrestricted optimizer on the supporting subfamily, hence has the contact-forced rows of Theorem A.15.1. Stopping before the two training preparations cannot occur with positive supporting probability: for each retained first-word branch, two possible unused second words force different strict vertices. A positive stopped component cannot belong to both vertices. The witnesses below therefore concern actual post-training validation cuts.

At a cut with two slots consumed from one tape and none from the other, the eight deterministic vertices and four disjoint additional faces from Theorem A.14.3 give twelve. That argument required only the four positive target diagonal words. It is still valid with full target support. Exchanging the two raw tapes negates the output and gives exactly the same channel-polytope certificate at the reversed cut. Thus either serial order needs twelve.

We give explicit certificates at the two possible nonserial entrance cuts. After one slot of each tape, write $x = X_1$, $y = Y_1$, and use suffix coordinates $(u, w, v, z) = (X_2, X_w, Y_2, Y_w)$. Define

i	$g_i^0(u, w)$	$g_i^1(u, w)$
1	$(1 - u)(1 - w)$	u
2	$1 - u$	u
3	$1 - u$	uw

and let $d_i^{xy} = g_i^y(v, z) - g_i^x(u, w)$. The twelve expressions with $i = 1, 2, 3$ and $x, y \in \{0, 1\}$ give ten distinct deterministic vertices. The only repetitions are $d_2^{11} = d_1^{11}$ and $d_3^{00} = d_2^{00}$, as is seen by substituting $u, w, v, z \in \{0, 1\}$ in the displayed functions.

There are four more required channels. Use count zero at prefixes $(x, y) = (0, 1), (1, 0)$ and count four at these same two prefixes. Their endpoints are respectively the continuations of (e_0, e_1) and (e_4, e_3) , with nontrivial weights $1 - t, t$. Each excludes simultaneous occurrence of the indifferent atom. The minimal face of each such channel consists of distributions supported, at each suffix, on the two endpoint output values. Among the ten mandatory vertices it contains only $d_1^{01}, d_1^{10}, d_3^{01}, d_3^{10}$, respectively. These four faces are pairwise disjoint. In their listed order, suffixes separating their six pairs are

pair	12	13	14	23	24	34
(u, w, v, z)	0110	0010	0000	0000	0001	0000

At each listed suffix the two sets of permitted output integers are disjoint; this follows directly from the displayed g_i^x and the masks e_0, e_4 . None of the required mixed rows is the sole mandatory vertex in its face, because $0 < t < 1$. Face packing gives $10 + 4 = 14$.

Suppose instead two slots of X and one of Y have been consumed. A deterministic continuation is indexed by (i, X_1X_2, Y_1) . The following twelve choices are distinct and include every strict-count continuation:

i	(X_1X_2, Y_1)
1	$(00, 0), (00, 1), (01, 0), (01, 1), (11, 0), (11, 1)$
2	$(00, 0), (01, 0)$
3	$(00, 1), (01, 1), (11, 0), (11, 1)$

They are verified by the formula $e_i(Y_1, Y_2, Y_w) - e_i(X_1, X_2, X_w)$ on the eight suffixes (X_w, Y_2, Y_w) . At training count zero take additionally prefixes $(01, 0)$ and $(11, 0)$. The corresponding two faces have endpoints $(0, (1 - Y_2)(1 - Y_w))$ and $(-1, -1 + (1 - Y_2)(1 - Y_w))$, respectively. Each contains just its listed $i = 1$ vertex, and they are disjoint at any suffix with $Y_2 = 1$. Both mixtures are nontrivial. Thus this cut also requires fourteen. Swapping X, Y proves the same result at the cut with one and two slots consumed.

Any nonserial interleaving either reads one slot from each tape first, or starts with two slots of one tape and then switches before its mark. It therefore passes one of the preceding cuts and needs fourteen. Every serial order has an upper realization of peak twelve: select the same five-valued frozen event and use its future residual functions. For full-support raw words the validation widths in either serial order are

$$(5, 10, 12, 10, 10, 7, 3).$$

These counts follow by restriction of the five masks, exactly as in Lemma A.11.4; for the reverse order negate all output residuals and exchange the tapes. Training and the charged stochastic selection cut have width at most five. Therefore both serial orders attain twelve, proving all assertions by Theorem A.8.2. The physical specialization follows from Theorem A.16.1. $\square$

This theorem optimizes the externally declared schedule; it does not optimize a report-dependent scheduler that changes which tape is read next. Such a scheduler has a different action interface, and its protocol state cannot be supplied for free. The phrase “all orders” in Theorem A.17.1 refers to all twenty declared interleavings, with this quantifier fixed explicitly.

Theorem A.17.2 (A singular support transition in reverse serial memory). *For every $\gamma \in [6/25, 13/50]$, the exact minimum peak in the reverse serial order $YYYXXX$ for the ideal target $Q_{\gamma,0}$ is ten. For every $0 < \kappa \leq 1/1000$ it is twelve. In particular the reverse-serial exact memory of the physical collision experiment jumps from ten at its zero-noise target to twelve at every positive $\sigma \leq 1/24$, although its target laws converge in total variation as $\sigma \downarrow 0$.*

Proof. For the upper bound at the ideal target, ignore its first report bit; its second bit equals the first almost surely, so this does not remove any needed information. After its second bit and mark, retain the residual $c - e_i(X)$, where $c = e_i(Y)$. Event e_2 has $c = 1$ surely; each other event permits two values of c . The compatible reverse validation profile is

$$(5, 5, 9, 9, 10, 7, 3).$$

After the second target bit the nine functions are, with the unread mark denoted by z ,

$$-e_0, 1 - e_0, 1 - z - e_1, 1 - e_1, 1 - e_2, 1 - e_3, z - e_3, 1 - e_4, -e_4.$$

Restricting after the target mark gives nine functions $c - e_i$. After the first candidate bit they give ten distinct functions of (X_2, X_w) ; after its second bit they give the seven scalar mark functions in Lemma A.11.4. Equality of residual functions is preserved by restriction, so these counts describe one compatible machine, not independent cutwise colorings. Training remains of peak five and the terminal score is correct almost surely for every candidate. On target words of zero probability, choose arbitrary legal transitions.

For the converse examine the cut after all three target slots and the first candidate bit. Write suffixes (X_2, X_w) in the order 00, 01, 10, 11. The three strict training events force the following eight distinct deterministic output channels:

$$\begin{aligned} (0, 1, 1, 1), & \quad (-1, 0, 0, 0), \quad (1, 1, 0, 0), \quad (0, 0, -1, -1), \\ (0, 0, 1, 1), & \quad (-1, -1, 0, 0), \quad (1, 1, 1, 0), \quad (0, 0, 0, -1). \end{aligned}$$

At training count zero take $X_1 = 0$ and target words 001 and 110. They force the channels with output means

$$(-t, 0, 0, 0), \quad (1 - t, 1, 1, 1).$$

Every integer coordinate is a point mass, and the remaining coordinate is the indicated mixture of neighboring output integers, because the target event indicator is fixed on these words. Their faces each contain exactly one of the eight mandatory vertices. They are disjoint at suffix 01, which requires output zero in the first and one in the second. Thus two additional generators are required, giving ten. All these prefixes have positive supporting probability. The full-support value twelve follows from the preceding theorem. Formula (84) gives the physical assertion and $\mu_1, \mu_2 \rightarrow 0$ by the uniform Gaussian bound. $\square$

The discontinuity concerns exact attainment. It neither contradicts continuity of the optimal score nor prevents a lower-memory device from approximating that score. It also explains why a

support-zero compiler cannot be transferred to the noisy physical target by simply keeping its state count unchanged.

APPENDIX A.18. PREPARATION LIMITS AND LOCALIZATION OF LEAST-FAVORABLE PRIORS

The contact structure has a second consequence, now for an arbitrary number of training preparations. We separate a general localization principle from the special geometry of the marked product family. An exact finite- N two-point prior is not assumed.

Theorem A.18.1 (Nearest-experiment limit and prior localization). *Let P_θ be a continuous family on a finite alphabet of size d , with compact metric parameter space, and let Q be fixed. Put*

$$d_Q(\theta) = \text{TV}(Q, P_\theta), \quad d_* = \min_{\theta} d_Q(\theta), \quad C_* = \{\theta : d_Q(\theta) = d_*\}.$$

For $N \geq 1$, the unrestricted frozen-validation values satisfy

$$(86) \quad d_* - \sqrt{(d-1)/N} \leq U_N(P, Q) \leq d_*, \quad U_N \uparrow d_*.$$

Every least-favorable prior μ_N in Theorem A.4.1 satisfies

$$(87) \quad \int (d_Q - d_*) d\mu_N \leq \sqrt{(d-1)/N}.$$

Consequently every weak limit of such priors is supported on C_ . For every $a > 0$ their mass on $\{d_Q - d_* \geq a\}$ is at most $\sqrt{(d-1)/N}/a$. More generally, an upper bound b_N on twice the expected total-variation error of a common empirical reconstruction replaces $\sqrt{(d-1)/N}$ in both conclusions.*

Proof. Let $\hat{P}$ be the empirical law of the N reports, and choose the indicator $h_{\hat{P}}$ maximizing $(Q - \hat{P})h$. If h_θ maximizes $(Q - P_\theta)h$, the two event inequalities give

$$(Q - P_\theta)h_{\hat{P}} \geq d_Q(\theta) - 2\text{TV}(\hat{P}, P_\theta).$$

This inequality holds for the same parameter-independent response rule at every parameter. For any law P on d atoms,

$$\mathbb{E} \text{TV}(\hat{P}, P)^2 \leq \frac{d}{4N} \left(1 - \sum_x P(x)^2 \right) \leq \frac{d-1}{4N}.$$

Thus its expected score is at least $d_Q(\theta) - \sqrt{(d-1)/N}$. Every response has score at most $d_Q(\theta)$ at that parameter, which proves the upper bound at a minimizing parameter. Ignoring the last observation proves monotonicity and hence the limit.

Against a minimizing prior, the Bayes optimum is U_N . It is at least the prior average of the same empirical rule. Therefore

$$U_N \geq \int d_Q d\mu_N - \sqrt{(d-1)/N}.$$

Combine this with $U_N \leq d_*$ to get (87). Markov's inequality proves the mass estimate. The function $d_Q - d_*$ is nonnegative and continuous on the compact parameter space, so its integral under any weak limit is zero. Such a limit is supported on its zero set. The proof uses only the common reconstruction bound, which proves the last assertion. $\square$

The empirical estimate in (86) is the same finite-alphabet mechanism underlying the retained decision-spectrum classification. Its role here is different: it controls every least-favorable prior, not merely a constructive selector. In particular, finite-preparation saddle supports cannot remain far from the closest candidate experiments as the preparation number grows.

Theorem A.18.2 (Intrinsic marked contact geometry). *For every $\gamma \in I$, set $s_\gamma = \sqrt{(1+\gamma)/2}$ and*

$$d_\gamma = 2s_\gamma - 1 - \gamma/2, \quad C_\gamma = \{(s_\gamma, s_\gamma), (1 - s_\gamma, 1 - s_\gamma)\}.$$

Then

$$(88) \quad \min_{p,q} \text{TV}(Q_\gamma, b(p, q)) = d_\gamma, \quad \operatorname{argmin}_{p,q} \text{TV}(Q_\gamma, b(p, q)) = C_\gamma.$$

Moreover, for Euclidean distance in $[0, 1]^2$,

$$(89) \quad \text{TV}(Q_\gamma, b(p, q)) - d_\gamma \geq \frac{7}{2300} \operatorname{dist}((p, q), C_\gamma)^2.$$

The constant is uniform on I and is derived from the experiment's atom probabilities, not from a metric-entropy or small-ball assumption.

Proof. Put $a = (1-p)(1-q)$, $c = pq$, $l = (1-\gamma)/2$ and $h = (1+\gamma)/2$. The common mass of target and candidate is $\psi(a) + \psi(c)$, where

$$\psi(z) = \begin{cases} z, & 0 \leq z \leq l, \\ z/2 + (1-\gamma)/4, & l \leq z \leq h, \\ 1/2, & h \leq z \leq 1. \end{cases}$$

This follows by summing the smaller mass on each of the two marked atoms corresponding to the report pair. Reflection $(p, q) \mapsto (1-p, 1-q)$ exchanges a, c ; assume $c \geq a$. If $c < 1/4$, both terms are in the first segment of ψ , and their sum is at most $1/2$.

For $c \geq 1/4$, put $x = \sqrt{c} \geq 1/2$. By the arithmetic-geometric mean inequality,

$$a \leq (1-x)^2 \leq 1/4 < l.$$

Consequently the common mass is at most $F(x) = (1-x)^2 + \psi(x^2)$, with equality only when $p = q = x$. On the three consecutive intervals with endpoints $1/2, \sqrt{l}, \sqrt{h}, 1$, this function is respectively

$$(1-x)^2 + x^2, \quad (1-x)^2 + x^2/2 + (1-\gamma)/4, \quad (1-x)^2 + 1/2.$$

The first two are strictly convex and the last strictly decreases. The first increases away from $1/2$. Thus a maximum must occur at $\sqrt{l}$ or $\sqrt{h}$. Their difference is

$$(90) \quad F(\sqrt{h}) - F(\sqrt{l}) = \gamma \left(\frac{3}{2} - \frac{2}{\sqrt{h} + \sqrt{l}} \right) > \frac{7}{575}.$$

Indeed $\sqrt{h} > 39/50$, $\sqrt{l} > 3/5$, and $\gamma \geq 6/25$, so the right side is greater than $(6/25)(3/2 - 100/69) = 7/575$. The unique maximizing diagonal point on this side is $x = \sqrt{h}$. Its common mass is $(1 - \sqrt{h})^2 + 1/2 > 1/2$, also excluding $c < 1/4$. Subtracting it from one gives d_γ . Reflection proves (88) and the assertion about all minimizers.

We record a uniform quantitative form. Write $s = \sqrt{h}$ and $\Delta = 7/575$. For $1/2 \leq x \leq \sqrt{l}$, the common mass is at most $F(\sqrt{l})$, so its deficit from $F(s)$ is at least Δ . For $\sqrt{l} \leq x \leq s$, convexity puts F below the chord between its endpoint values. Since this interval has length at most one, its deficit is at least $\Delta(s-x)$. For $s \leq x \leq 1$ the deficit is

$$(1-s)^2 - (1-x)^2 = (x-s)(2-s-x) \geq (1-s)(x-s).$$

Since $s < 4/5$, this too is at least $\Delta|x-s|$. Together the three intervals give $F(s) - F(x) \geq \Delta|x-s|$.

Let $z = (p-q)^2$. At fixed $c = x^2$ the additional loss from the nondiagonal candidate is exactly

$$(1-x)^2 - a = p + q - 2x = \frac{z}{p+q+2x} \geq z/4.$$

The geometric mean x lies between p and q , and hence

$$(p - s)^2 + (q - s)^2 \leq 4z + 4(x - s)^2 \leq 4z + 4|x - s|.$$

It follows that the total deficit is at least $(\Delta/4)((p - s)^2 + (q - s)^2)$, proving (89) in this case. If $c < 1/4$, the total variation is at least $1/2$, whereas $d_\gamma < 12/25$ from $s < 4/5$ and $\gamma \geq 6/25$. The excess is therefore greater than $1/50$; Euclidean squared distance in the square is at most two, and $1/50 > 2(7/2300)$. This proves the remaining case. Reflection completes the proof. $\square$

Corollary A.18.3 (An all-preparation law and a limiting two-point prior). *For all $N \geq 1$ and $\gamma \in I$,*

$$(91) \quad d_\gamma - \sqrt{3/N} \leq U_N(b, Q_\gamma) \leq d_\gamma, \quad U_N(b, Q_\gamma) \uparrow d_\gamma.$$

Every least-favorable prior $\mu_{N,\gamma}$ satisfies

$$(92) \quad \int \text{dist}(\theta, C_\gamma)^2 d\mu_{N,\gamma}(\theta) \leq \frac{2300}{7} \sqrt{3/N}.$$

A least-favorable prior can be chosen invariant under simultaneous complementation of the two parameters. Every such symmetric choice satisfies, for the Euclidean quadratic transport distance,

$$(93) \quad W_2 \left(\mu_{N,\gamma}, \frac{1}{2} \delta_{(s_\gamma, s_\gamma)} + \frac{1}{2} \delta_{(1-s_\gamma, 1-s_\gamma)} \right) \leq \sqrt{2300/7} (3/N)^{1/4}.$$

In particular, the limiting symmetric least-favorable law is unique.

Proof. Estimate the four-atom report-pair law from training and tensor it with the known fair mark. Tensoring with a common probability preserves total variation. The proof of Theorem A.18.1 therefore uses $d = 4$ rather than eight, proving (91) and an excess bound $\sqrt{3/N}$. Combining the latter with (89) gives (92).

Simultaneously complement both report bits and the mark in every training and validation word. This leaves Q_γ invariant and sends $b(p, q)$ to $b(1 - p, 1 - q)$. The Bayes objective in (25) is invariant under this transformation and convex in the prior. Averaging a minimizing prior with its reflected image is therefore again minimizing and is symmetric.

Map each parameter to a closest point of C_γ , choosing the two points with equal probabilities at a tie. This is a Borel kernel, equivariant under reflection. A symmetric prior is pushed forward to the equal two-point law in (93); the induced coupling has squared transport cost exactly the left side of (92). Taking square roots proves the assertion. $\square$

These conclusions identify the endpoint and limiting contact geometry for every preparation number. They do not assert that the exact prior has two atoms at every finite N , or that the displayed convergence rates and constants are sharp. The finite- $N = 1, 2$ saddles are preserved as exact results. The new theorem explains their relationship to an intrinsically determined limiting contact set without extrapolating their particular event tables to unproved values of N .

APPENDIX A.19. AN EXACT PREPARATION–DECISION–MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training–validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh

independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem A.19.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(94) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(95) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (94).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary A.19.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(96) \quad N_c^{\text{er}}(K) = \left\lfloor \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rfloor + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (96), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

APPENDIX A.20. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem A.19.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(97) \quad B_{m,K} = 1 - \frac{(K - s)q^2 + s(q + 1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition A.20.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(98) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m + s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (95) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q + 1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - |\text{fiber}|/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(99) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (99) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil/m$. This proves the last assertion. $\square$

Theorem A.20.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor/m$ and $b = \lceil m/2 \rceil/m$. Then*

$$(100) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m + 1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition A.20.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead

freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (98) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section A.6 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

APPENDIX A.21. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r-1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(101) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma A.21.1 (Causal interpretation of the coefficients). *Every coefficient in (101) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (101) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at

most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem A.21.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(102) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (101) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma A.21.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (102). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark A.21.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem A.4.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

APPENDIX A.22. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma A.22.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-}(L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem A.22.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(103) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(104) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma A.22.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (104).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_{\pm}^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$.

Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_{\pm}^n(i, \cdot), \pi_{\pm}) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (104). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (103). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (104) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

APPENDIX A.23. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary A.23.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\overline{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding’s inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition A.23.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (25); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

APPENDIX A.24. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem A.24.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\overline{D}_R > c$, have error bounds

$$(105) \quad \sup_{\text{private}} \mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(106) \quad \sup_{\text{equal}} \mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem A.2.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (105)–(106) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary A.12.2 and Proposition A.12.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding's inequality proves (105). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (106) by the same inequality. Retain the integer sum in

$\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem A.2.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem's stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

APPENDIX A.25. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section A.21 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section A.22 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (62), polynomial identity (64), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

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